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SURVEY

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Teacher-Student Framework for Polyphonic Semi-supervised Sound Event Detection: Survey and Empirical Analysis

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Polyphonic sound event detection refers to the task of automatically identifying sound events occurring simultaneously in an auditory scene. Due to the inherent complexity and variability of real-world auditory scenes, building robust detectors for polyphonic sound event detection poses a significant challenge. The task becomes furthermore challenging without sufficient annotated data to develop sound event detection systems under a supervised learning regime. In this article, we explore the recent developments in polyphonic sound event detection, with a particular emphasis on the application of Teacher-Student techniques within the semi-supervised learning paradigm. Unlike previous works, we have consolidated and organized the fragmented literature on Teacher-Student techniques for polyphonic sound event detection. By examining the latest research, categorizing Teacher-Student approaches, and conducting an empirical study to assess the performance of each approach, this survey offers valuable insights and practical guidance for researchers and practitioners in the field. Our findings highlight the potential benefits of utilizing multiple learners, ensuring consistent predictions, and making thoughtful choices regarding perturbation strategies.

CCS Concepts: • **Computing methodologies** → **Neural networks**; *Semi-supervised learning settings*;

Additional Key Words and Phrases: Polyphonic sound event detection, Teacher-Student framework, semi-supervised learning

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1 Introduction

Navigating our day-to-day environment relies on how we perceive and interpret our surroundings. Our perception of sceneries is influenced by two main cues: visual and auditory [94]. The auditory cues are complex mixtures of sound events such as *water tap running, smartphone notifications ringing, birds chirping,* and *people chatting*. Our auditory system is naturally capable of detecting, identifying, and locating sound events with the objective of soliciting the appropriate response. Considerable efforts have been undertaken with the aim of developing computational systems that can emulate the functions of the human auditory system. The evolution of such systems has led to

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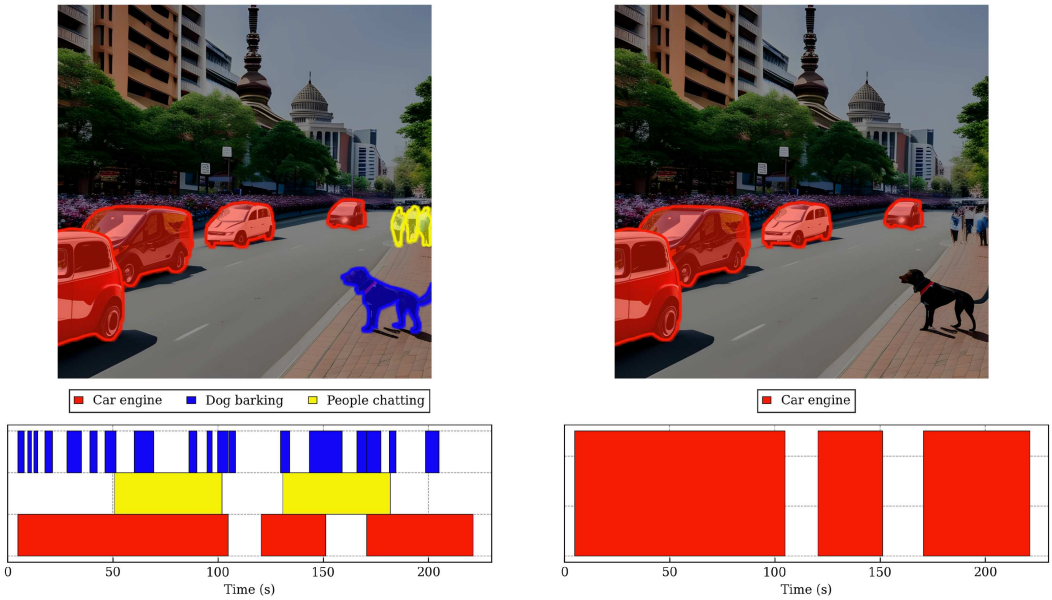


Fig. 1. Comparison of polyphonic and monophonic SED systems. (a) A polyphonic sound scene with three different sound events of interest. (b) A monophonic sound scene with a single sound event of interest.

the emergence of several, distinct, but complementary tasks that fall under the umbrella term of **Computational Auditory Scene Analysis (CASA)** [143]. In this survey, we shed light primarily on the **Sound Event Detection (SED)** task. Our aim is to provide the reader with insights on the most recently observed trends in **Polyphonic Sound Event Detection (PSED)**.

SED refers to the task of recognizing the types of events occurring in an auditory scene; *what is happening?* and pinpointing the respective temporal start and end times of these occurrences; *when is it happening?* As shown in Figure 1, sound event detectors can be categorized into two major categories: *monophonic* and *polyphonic* [94]. At a given time, a monophonic system is designed to only detect a singular event, whereas a polyphonic system is capable of detecting more than one event [143]. In this work, we focus on PSED systems by virtue of it being more in line with realistic scenarios, as sounds seldom occur in isolation.

The interest in SED is largely fueled by the resonating impact of automating the aforementioned task in a wide range of applications [143]. These applications span across a large number of research fields where sound sensing is involved. Sound detection is of interest in fields where sound sources are *anthrophonic*, such as the work of Varghees and Ramachandran on heart sounds and murmurs [102], or *biophonic*, such as the work of Lostanlen et al. on avian flight calls [86]. A plethora of strategies have emerged for the development of sound event detectors, regardless of the nature of sounds and the final application domain. Among these strategies, **Supervised Learning (SL)** is the dominant approach for tackling the SED task [94]. For deep learning models to be fully operational, under the supervised paradigm, it is necessary that an extensive, manually annotated dataset be available [143]. However, the annotation procedure is costly, laborious, and is likely disposed to labeler subjectivity [94]. As a result, researchers have branched out toward **Semi-Supervised Learning (SSL)** to alleviate the hampering need for reference annotations [21]. Semi-supervised algorithms exploit extensive amounts of unlabeled data, in conjunction with small sets of labeled

data, to train sound detectors [111]. Most recently, noticeable interest has been assigned to study semi-supervised Teacher-Student techniques, which are methods that encourage two acoustic models, a *student* and a *teacher*, to generate consistent predictions [151].

In this survey, we aim to offer the reader with an overview on Teacher-Student techniques for PSED. The favorable performance of Teacher-Student-based detectors during the recent editions of the **Detection and Classification of Acoustic Scenes and Events (DCASE)** challenges served as the primary motivation for conducting this survey. To the best of our knowledge, and at the time of writing this survey, the literature surrounding Teacher-Student techniques for SED have not yet been collected and organized in one record. To substitute for the lack, we provide explanations and algorithms that summarize the internal mechanics of all methods under review. To guide future practitioners, we include a categorization of Teacher-Student methods that illustrate the manner in which these methods relate to one another and within the scheme of SSL. Lastly, we expand our work with an empirical study in which we test the inner workings of each surveyed method to assess its efficiency and applicability.

The remainder of this article is organized as follows. Section 2 further delves into additional motivations that prompted us to undertake this survey. In Section 3, we formulate the SED problem and illustrate how the task is carried out. In Section 4 and Section 5, we describe the classical and the more contemporary approach for SED, respectively. We dive into the common characteristics of the surveyed methods in Section 6 and describe each technique in Section 7. In Section 8 and Section 9, we present our experimental setup and lay out our empirical findings. Finally, in Section 10, we provide prospects for the potential future of semi-supervised audio event detection.

2 Motivation and Prior Work

The critical need for event detection in audio streams has prompted researchers to extensively study the task in recent years. An increase in research activity is noticeable through the multitude of evaluation campaigns dedicated to the task, which include the **Classification of Events Activities and Relationships (CLEAR)** campaigns, namely: CLEAR2006 [64], CLEAR2007 [65], and DCASE challenges from 2016 through 2021 [63, 89, 93, 109, 111, 138]. With the substantial amount of attention geared toward SED, a plethora of algorithms, methods, and techniques have emerged. Nonetheless, very few attempts have considered reviewing the SED-related literature. The earliest attempt at reviewing SED systems was published by Stowell et al. in 2015 [130]. The article covers various approaches used for feature extraction for SED and discusses different algorithms used for the detection of sounds including threshold-based methods and machine learning-based methods. Through their work, the authors underline the importance of machine listening tasks and the need for more sophisticated algorithms as well as more robust evaluation metrics.

Another notable attempt at reviewing the state-of-the-art of SED is the work published by Babaee et al. in 2017 [9]. The authors reviewed a total of 66 different articles related to the detection and classification of events that have been published from early 2000s to 2014. The review is relatively expansive but can be deemed outdated due to the speed at which literature on SED has grown. Since the publication of Babaee et al., many advances in the field of auditory analysis have been made, and many novel approaches for SED have been proposed; most notably, the advent of deep learning-based methods that were placed at the forefront of studies on SED system development.

In 2017, Dang et al. published a short survey on deep learning techniques applied to PSED [30]. The authors provide an overview of the challenges associated with PSED and the various deep learning models that have been developed to address these challenges. Additionally, the paper covers the evaluation metrics put in place for performance assessment as well as the different datasets used for training and testing PSED systems.

Xia et al. later contributed to the literature with a comprehensive survey on **Acoustic Event Detection (AED)** in 2019 [147]. Specifically, the authors focused on the deep learning-based approaches for both strongly labeled and weakly labeled SED systems, with an emphasis on the proficiency of deep **Neural Networks (NNs)**. The authors highlighted the advantages of NN-based deep learning approaches over conventional methods when applied to SED.

In 2020, Chan and Chin contributed to the existing literature with a review of different PSED systems with an emphasis on the employed learning architectures [20]. The authors categorized the surveyed systems into non-NN-based and NN-based methodologies. The NN-based methods are further categorized into non-hybrid models, hybrid models, and models built using weakly labeled data. The review provides in-depth discussions on each work and thorough examinations of the limitations of each method.

The most recent publication gathering related literature and advances on SED is the work of Mesaros et al. published in 2021 [94]. This work encompasses all aspects associated with the task thoroughly and acts as an up-to-date guide for new practitioners and adepts alike. The authors provide clear descriptions of the challenges facing the SED community, machine learning approaches, datasets and data-related issues, NN architectures, and performance metrics. The article also discusses related research problems regarding augmentation methods, feature extraction techniques, alternatives to data collection processes, and the main differences between AED and other audio-related tasks. The tutorial is relatively comprehensive and treats all building blocks of SED. However, it does not detail the specifics of each concept but rather seeks to provide the reader with a general overview on the whys and wherefores of each concept.

The aforementioned reviews and surveys cover the literature surrounding SED in a broad and general scope, focusing primarily on early supervised-based methods. However, the most recent advances in SED are associated with the use of large pools of unlabeled data SSL methods. Specifically, the adoption of Teacher-Student techniques has recently been extensively used for this task. This trend can be portrayed by the large body of publications and submissions in the recent DCASE editions [111]. We believe that the success of these techniques at semi-supervised event detection is an indication of the power of unlabeled signals for learning tasks. Therefore, a deeper dive into the internal mechanics of these techniques is of paramount importance, as it will guide current and future research endeavors in the pursuit of ever more proficient SEDs. It is worth noting that we do not intend to cover all existing Teacher-Student methods in literature. Rather, this work covers Teacher-Student techniques that have been applied to PSED *only*. Specifically, we cover the literature surrounding three techniques, namely, Mean Teacher, Guided Learning, and Interpolation Consistency Training.

3 Problem Formulation

An event refers to a discrete and identifiable sound produced by a physical entity and is typically characterized by a specific duration in time [143]. Let us consider the example of a wildlife recording. An event in this context refers to any unit of sound that is distinguishable within a mixture of sounds, such as *bird calls* and *mammal vocalizations*. The goal of SED is to identify and isolate these events for subsequent use in various applications. Given a raw waveform, the feature extraction phase transforms the signal into a compact representation of the audio content. The goal is to capture the most relevant and discriminative information from the audio signal that can help distinguish between different sound events [143]. To achieve this, the signal is first fragmented into T analysis frames of fixed length, which generally ranges from 20 to 60 ms. Next, the appropriate computations are applied, frame by frame, to produce the chosen acoustic features. It is worth noting that, the choice of the feature extraction method is very often *heuristically decided* [143]. Nonetheless, the most prominent features currently used for SED include **Mel Frequency Cepstral Coefficients (MFCC)** and log-Mel energies [9]. Let $X \in \mathbb{R}^{T \times F}$, be the feature matrix, where F

represents the number of extracted acoustic features. Given a strongly labeled dataset, each feature vector $x_t = [x_t^1, x_t^2, ..., x_t^F] \in X$, where $t = 1, 2, ..., T$, is associated with a target vector $y_t \in Y$, where $Y \in \mathbb{R}^{T \times C}$ and C denotes the number of predefined sound event classes. Each target in Y is a binary vector that represents the presence or absence of each event class with 1 signaling the presence of the event. The goal of the detector is to learn a mapping function $f : X \longrightarrow Y$ that best represents the relationship between the input signals and the target annotations. Given a feature vector x_t, the acoustic model outputs a vector $\hat{y}_t = [\hat{y}_t^1, \hat{y}_t^2, ..., \hat{y}_t^C]$, where $c \in C$, corresponds to the frame-level class presence probabilities within the time analysis frame. To obtain the timestamps of each event occurrence, i.e., detect the events, the predictions must be post-processed. Post-processing usually involves two stages: a soft-label and a hard-label stage [37]. Given a vector $\hat{y}_t$ of predicted class probabilities, soft-label post-processing transforms a per event probability estimate $\hat{y}_t^c$ into a binary representation. This is generally performed via *thresholding* over a constant σ as follows:

$$\hat{y}_t^c = \begin{cases} 1 \text{ if } \hat{y}_t^c > \sigma \\ 0 \text{ otherwise} \end{cases}$$

During the hard-label post-processing stage, the binary sequences are smoothed out to remove detection artifacts and noise such as false positives and false negatives. A popular choice for smoothing the predictions is *median filtering* [37]. A median filter operates by traversing the signal in a sliding window manner and calculates the median value within each window of size ω. Hard-label post-processing methods also include double thresholding [37] and short duration event removal and accept gaps [52].

In a classical supervised setting, the acoustic model learns from a corpus of strongly labeled audio clips $D_L = (X_i', Y_i')_{i=1}^N$, where N is the total number of clips in the training set. Due to a multitude of labeling-related constraints, recent works on PSED have incorporated sets of unlabeled audio clips to the training set [151]. We denote an unlabeled collection of clips as $D_U = (X_i')_{i=1}^M$, where M is the number of unlabeled instances. The training set in a semi-supervised setting can thus be denoted as $D = D_L \cup D_U$. It is worth noting that in the majority of SSL-based SED cases, the number of unlabeled training examples is substantially greater than labeled examples; put formally, $M \gg N$. The premise behind incorporating large unlabeled sets in SSL techniques is to build sound event detectors that exceed in performance their counterpart supervised detectors [151].

4 Supervised SED

Automatic SED is found to be analogous to **Automatic Speech Recognition (ASR)**, whereby each event in an audio stream can be considered a word in continuous speech. Inherently, it has drawn upon the foundations laid by ASR to differentiate between different sound sources and identify patterns in audio data [143]. Much like ASR, the earliest sound event detectors were built using conventional statistical models such as **Hidden Markov Models (HMM)** and **Support Vector Machines (SVM)**.

In 2006, the **AED and Classification (AED/C)** evaluations were carried out as part of the CLEAR campaign [64]. Three participant partners from the **Computers in the Human Interaction Loop (CHIL)** project were evaluated using two databases of isolated acoustic events and a database of interactive seminars. The results indicated that the best-performing AED system was based on an HMM with a left-to-right topology. The AED/C evaluation provided a valuable benchmark for evaluating the performance of automatic SED systems. Most importantly, it highlighted the importance of using standardized evaluation methodologies and objective metrics and helped to advance the state-of-the-art in this field.

In 2013, researchers from the **Tampere University of Technology (TUT)** and the University of Surrey organized a challenge aimed at evaluating AED and AEC systems, which culminated in a

series of annual challenges known as the DCASE challenges. In DCASE2013, the SED subtask was designed to address two challenges: detecting events without overlapping sounds and detecting events in a polyphonic scenario [43]. The baseline system was based on a **Non-negative Matrix Factorization (NMF)** framework, which has previously been shown to be pertinent at modeling the spectral characteristics of audio sources in an acoustic mixture. The challenge attracted a total of 10 participants and over 10 systems were submitted for evaluation. The results of the challenge showed that the challenge baseline was outperformed by the majority of systems. The evaluation proved that variants of HMMs were reliable tools for event detection, due to their ability to model timeseries data. Table 1 provides a summary of SED methods using traditional learning techniques. The collection of works in Table 1 suggests that the dominant approach to supervised SED, in its infancy, relied on models such as SVM [87], [140], HMM [28], [160], and Gaussian Mixture Models [41], [163]. These systems were typically trained on handcrafted features such as MFCC and **Linear Predictive Coding (LPC)**, in addition to time-domain features such as **Zero Crossing Rate (ZCR)**.

Despite the strong performances achieved by the systems mentioned above, traditional machine learning models were found to suffer from sensitivity to noise and reverberation and have difficulty modeling long-term dependencies. The latter limitations render the task of capturing the temporal evolution of longer sound events especially difficult. To overcome the aforementioned setbacks, researchers have been exploring the use of end-to-end deep learning models for SED, which can learn both the feature extraction and classification components in a single network [94].

Çakir et al. issued a notable empirical study on the performance of **Convolutional Recurrent NN (CRNN)** in PSED and showed a substantial improvement in performance over other established methods [16]. Challenges following DCASE2013 have been instrumental in demonstrating the potential of deep learning architectures for SED. The submissions provide evidence on the utter shift from traditional shallow models to complex deep networks. Table 2 lists various works on supervised deep learning-based systems employed in the SED task, including the data corpus, the architecture, and the feature extraction methods. The adoption of deep learners has unequivocally reduced the need for handcrafted and domain expertise-based feature extraction methods; hence, the overwhelming majority of works are based on spectrograms and variants thereof. Since deep learning models require a large amount of training data, training procedures for SED have included a data augmentation step in the overall pipeline. Data augmentation has been used to artificially increase the size of the training set by generating new examples through adding noise, changing the pitch or speed, or applying other transformations to simulate different recording conditions.

The works mentioned in Table 2 have tackled automatic SED following an SL approach. While SL methods have demonstrated satisfactory outcomes, they exhibit several limitations, the most critical of which is the need for annotations [94]. Under this paradigm, SED systems learn to classify and to timestamp events from a strongly annotated dataset, i.e., a dataset containing strong labels. A strong label refers to a description of an audio excerpt that includes the temporal information of each event instance in addition to the identity of the instances [94]. Numerous approaches to obtain strong labels are available in practice such as manual labeling and synthetic generation of audio signals [94]. Synthetic generation allows for the production of accurate labels without the need for human intervention or domain expertise. The process is carried out by audio mixing algorithms. Despite being a reliable method for generating strong labels, it is not infallible, since synthetically generated data may not be consistent with real-life sounds. On the other hand, manual labeling of audio recordings involves human intervention and can be done by expert annotators or via crowdsourcing. Nonetheless, a great number of factors can hinder the acquisition and the usage of manually labeled data in practice. A few of these factors include the following:

— Availability: Manually labeling audio recordings can be a remarkably laborious and time-consuming task [151]. It requires the annotator to repeatedly listen to an audio excerpt to

Table 1. Summary of Supervised SED Systems Using Traditional Methods

Reference	Datasets	Input Features	Learning Architectures
Niessen et al. [101]	IEEE AASP CASA OL [42]	STE, SF, MFCC, ZCR, SRoll-off, LPC, Flatness, Brightness	HMM, Random Forest
Chauhan et al. [22]	IEEE AASP CASA OL [42]	SC, SF, SE, STE, SRoll-off, MFCC, Loudness, WDC, Autocorrelation	Generalized Likelihood Ratio Test Classifier
Zhuang et al. [163]	Netcarity Fall [59]	Perceptual Linear Prediction	Gaussian Mixture Model, SVM-Gaussian Mixture Model supervector
Kumar et al. [72]	TRECVID 2011 [8]	MFCC, F-vector, GMM-MAP	Gaussian Mixture Model, Random Forest
Socoró et al. [128]	Dynamic Acoustic Mapping Corpus [153]	MFCC	Discriminant Analysis, Gaussian Mixture Model, SVM, k-NN, Universal Gaussian Mixture Model
Cotton et al. [28]	FBK-IRST Database [28]	MFCC, ConvNMF	HMM
Lu et al. [87]	Personal Corpus	Sparse Representations	Linear SVM
Zhou et al. [160]	CLEAR 2007 AED Evaluation [65]	Delta-MFCC, MFCC, Log-Frequency Filterbanks	One-pass HMM
Zhuang et al. [164]	CLEAR 2007 AED Evaluation [65]	MFCC, Log-Frequency Filterbanks	Tandem Connectionist HMM, SVM-Gaussian Mixture Model supervector
Schröder et al. [122]	IEEE AASP CASA OL [42], CLEAR2007 AED Evaluation [65], NOISEX-92 [139]	Mel-spectrograms, Gabor Filterbanks	Left-to-Right HMM
Geiger et al. [41]	Personal Corpus (DCASE 2013 [43], FreeSound [43])	MFCC, Gabor Filterbanks	Gaussian Mixture Model
Vavrek et al. [140]	Recorded Corpus of Gunshots	MFCC	SVM
Huang et al. [57]	TRECVID 2010 [105], TRECVID 2011 [8], TRECVID 2012 MED [106]	I-Vector	HMM, SVM

AASP, Audio and Acoustic Signal Processing; AED, Acoustic Event Detection; CASA, Computational Auditory Scene Analysis; CLEAR, CLassification of Events, Activities and Relationships; ConvNMF, convolutional non-negative matrix factorization; FBK-IRST, Fondazione Bruno Kessler - Istituto per la Ricerca Scientifica e Tecnologica; GMM-MAP, Gaussian Mixture Model-Maximum *A Posteriori*; IEEE, Institute of Electrical and Electronics Engineers; LPC, linear predictive coding; MED, Multimedia Event Detection; MFCC, Mel Frequency Cepstral Coefficients; OL, Office Live; SC, Spectral Centroid; SE, Spectral Entropy; SF, Spectral Flux; SRoll-off, Spectral Roll-off; STE, Short-Time Energy; TRECVID, TREC Video Retrieval Evaluation; WDC, Weighted Difference of Cepstral Coefficients; ZCR, Zero Crossing Rate.

Table 2. Summary of Modern Supervised SED Systems Using Deep Learning Architectures

Reference	Datasets	Input Features	Deep Architectures
Zhang et al. [155]	TAU Spatial Sound Events 2019 [5]	Log-Mel Spectrograms	Multi-Scale CRNN
Çakir et al. [16]	TUT-SED Synthetic 2016 [93], TUT-SED 2009 [16], TUT-SED 2016 [92], CHiME-Home [40]	Log-Mel Band Energies	Convolutional NN, Recurrent NN, CRNN
Cao et al. [17]	TAU-NIGENS Spatial Sound Events 2021 [110]	Mel Band Energies	SELDNet
Imoto et al. [58]	TUT-SED 2016 [93], TUT-SED 2017 [90], TUT-ASC 2016 [93], TUT-ASC 2017 [90]	Log-Mel Band Energies	Convolutional NN-Bi-GRU
Pérez López et al. [113]	TAU Spatial Sound Events 2019 [5]	Spectrograms	CRNN
He et al. [53]	TAU-NIGENS Spatial Sound Events 2021 [110]	MaxCorrelation Filterbanks	SoundDet
Jin et al. [61]	TUT-SED 2016 [92]	Log-Mel Band Energies	Capsule NN
Choi et al. [26]	TUT-SED Synthetic 2016 [93]	Mel-Band Energies	Deep NN
Kong et al. [69]	TUT-SED 2016 [92]	Mel-Filterbanks	Deep NN
Gorin et al. [47]	TUT-SED 2016 [92]	Log-Mel Filterbanks	Convolutional NN
Bastien et al. [11]	TUT-SED 2016 [92]	MFCC	Deep NN
Cao et al. [18]	TAU Spatial Sound Events 2019 [5]	Log-Mel Band Energies, Log-Mel Intensity Vector, GCC-PHAT	CRNN
Zöhrer et al. [165]	TUT-SED 2016 [92]	MFCC, Log-Magnitude Spectrograms	GRNN
Adavanne et al. [3]	TUT-SED 2016 [92]	Log-Mel Band Energies, Dominant Pitch Value and Periodicity	Recurrent NN-Long Short Term Memory
Adavanne et al. [4]	TUT Sound Events 2018 ANSYN [92], TUT Sound Events 2018 RESYN [92], TUT Sound Events 2018 REAL [40], TUT Sound Events 2018 REALBIG [40], TUT Sound Events 2018 CRESYN [92]	Spectrogram	CRNN

(Continued)

Table 2. Continued

Reference	Datasets	Input Features	Deep Architectures
Lasseck [75]	DCASE Bird Audio Detection Dataset 2018 [131]	Spectrogram	Convolutional NN
Vesperini et al. [142]	DCASE Bird Audio Detection Dataset 2018 [131]	Log-Scaled Mel-Spectrogram	Capsule NN
Kaiwu et al. [62]	TUT Rare Sound Events 2017 [90]	Spectrogram	Region-Based Fully Convolutional Network

ASC, Acoustic Scene Classification; CHiME, Computational Hearing in Multisource Environments; DCASE, Detection and Classification of Acoustic Scenes and Events; GCC-PHAT, Generalized Cross-Correlation with Phase Transform; GRNN, Gated Recurrent Neural Network; TAU, Tampere University; TUT-SED, Tampere University of Technology - Sound Event Detection.

adjust the label and time boundaries. Consequently, the size of strongly labeled datasets is usually limited to a few hours of audio data.

— Reliability: Manually annotating audio data can be, in many cases, erroneous. A noteworthy study has been conducted by Cartwright et al. in 2017 to quantify the reliability of crowd-sourced annotations [19]. The experimental results indicate that human annotations can be affected by the lack of expertise of the annotator, the complexity of audio signals, and the provided sound visualization aid. Erroneously annotated data can create noise for any learner and might lead to degraded performance.

In recent years, a considerable amount of research work has been devoted to finding better suited approaches to SED, in an attempt to bypass the current limitations of SL methods. As a result, the overwhelming majority of works on SED have branched out of the SL paradigm and into the SSL paradigm. The core notion of this on-going research topic relies on using limited amounts of strongly labeled data in conjunction with unlabeled data during the training stage.

5 Semi-Supervised SED

Despite the promising results of the supervised methodology to build SED systems, it is crucial to note that one fundamental specification for SL is the requirement for labeled data. This places a high demand for human and hardware resources, which inevitably puts the supervised method in confrontation with a fair number of weaknesses and renders the annotation process extremely difficult. One example of a tedious annotation procedure is the experimental setup of Mesaros et al. in 2010. The setup was specifically designed for the task of AED in real-life environments [91]. The training corpus was a collection of long-duration recordings going as far as 30 minutes per recording. All events within the recordings were manually annotated. Each recording context contains a minimum of 9 and a maximum of 16 different event classes, totaling to 61 event categories in the dataset. It is worthwhile mentioning that for a single context within the dataset, the number of event occurrences manually annotated can exceed 1,000.

Another example of similar laborious procedures is the renowned DCASE challenges. For the first edition of the DCASE challenges, several participants were recruited to perform pre-scripted audio events, such as *laughter, page turning,* and *mouse click* [43]. To record the scripts, a Soundfield microphone system was used to capture four-channel audio files. For every script, multiple takes were recorded, and the choice of the best take fell upon the one having the least amount of

unscripted noise; put differently, a manual evaluation of each take was necessary for this selection. This procedure resulted in 24 recordings of individual sounds per class (with a total of 16 classes). To annotate the recordings, the annotators were first trained to use the Sonic Visualiser software. Given the ambiguity of annotating sound events, two annotations were created per recording, which were later examined for consistency.

In 2016, the collective efforts of members of TUT led to the release of the TUT sound events 2016 dataset [93]. Each sample within the dataset was recorded in a different location to ensure a high level of variability in each acoustic category. The annotation of the recordings was done by two research assistants, who were previously trained on a few example recordings. Due to the polyphonic nature of the recordings, each excerpt had to be listened to a number of times during the annotation, which ultimately renders the procedure slower. For privacy-related issues, all recordings were screened for any privacy infringing fragment as well as any microphone failures or distortions.

The aforementioned examples provide insights on the laborious task of collecting and annotating data for SED purposes. Nonetheless, we can speculate that the collection procedure for everyday life sounds can be considered relatively uncomplicated, as the scenarios can be controlled and the annotations can be easily provided and verified by non-expert individuals. However, the task is found to be more tedious and resource hungry in a biological environment such as wildlife. In 2017, Stowell et al. designed a public evaluation campaign focused on the bird detection task [131]. To this end, the organizers gathered and annotated multiple datasets from various remote sources. Although a few of the source material included some form of metadata related to the labels, it was found to be insufficiently precise to be considered ground truth. Therefore, an annotation procedure was designed to label the data for the purposes of the evaluation. Most recordings were single annotated and sometimes double annotated, which allowed for inter-rater reliability estimates. The annotation was carried out by experienced listeners using headphones and a helper web interface.

To further illustrate the delicacy of recording and annotating bioacoustics material, we cite the bioacoustics event detection challenge included in DCASE2021 [98]. The annotation of the recordings within the dataset calls for a domain expert in bird vocalizations. Andrew Farnsworth, an expert ornithologist, has annotated all the audio data for the presence of flight calls from migratory passerines and has found 2,662 calls from 11 different species. The training set also included vocalizations of other mammals and birds such as spotted hyenas, jackdaws, and meerkats. It is worthy of note that recording and annotating vocalizations of animals in the wild require specific hardware resources, a meticulously studied recording process, and extensive domain expertise. The aforementioned examples demonstrate the challenges hindering the development of reliable sound event detectors, which can be summarized as follows:

- Data collection: Acquiring and storing audio data involves selecting appropriate recording equipment, identifying relevant sound sources, and capturing diverse soundscapes that cover a wide range of acoustic environments.
- Data annotation: Annotating data involves curating annotation guidelines and appropriate tools as well as ensuring the quality of the annotations. This can render annotation difficult, especially when the sound events are complex or subjective.
- Hardware and software resources: This involves selecting and configuring appropriate audio processing equipment and managing the resources effectively. It can be difficult to optimize the use of resources, especially when the task requires significant computing power or specialized hardware.
- Quality assurance: It involves developing evaluation procedures, measuring model performance against relevant benchmarks, and identifying sources of error or bias. It can be difficult

to ensure the quality of SED models, especially when the models are complex or operate in dynamic and uncertain acoustic environments.

— Ethical considerations: Potential ethical issues, such as privacy violations, bias, and discrimination, must be addressed. This involves ensuring that the audio data are collected and used ethically without infringing on the rights of individuals or communities. It can be difficult to balance the benefits and risks of SED technology, especially when the technology has the potential to impact the privacy and well-being of individuals.

The above-mentioned issues have posed a critical dilemma that has given rise to a potent interrogation: *how to collect and annotate larger amounts of data more efficiently?* In response to this, practitioners have gravitated toward diminishing the need for supervised signals via employing unlabeled data, i.e., no supervision signal. Put differently, the research community is currently unveiling the hidden potential of incorporating unlabeled data to learning processes, rather than continuing to harvest labeled data [94]. The plethora of currently available multimedia outlets have facilitated the access to abundant and ever so growing pools of audio/video material. This alleviates the costly annotation procedure and offers practitioners both availability and variability [21]. With the latter arises a new research question: *How to efficiently use unlabeled data to build powerful learners?* In response to this, research in SED have incorporated large sets of unlabeled data to train SED systems through a much more complex learning scheme: SSL.

SSL is a machine learning paradigm that lies between the supervised and the unsupervised paradigm. It has recently been extensively used in academia and industry for application domains that suffer from scarcity of labeled data such as SED. SSL methods can be categorized based on the final target use case into one of two schemes: inductive or transductive learning [94]. In inductive learning, the goal is to construct a general rule $f : X \rightarrow Y$ which can then be applied to predict labels for previously unseen test cases, i.e., unobserved instances at training time. In transductive learning, the rule f is defined to label the data within the use case, i.e., the observed instances at training time. In the context of this survey, we are only interested in inductive SSL. More specifically, we focus on the use of deep learning-based techniques, since these have proven succesfull at leveraging abundant supplies of unlabeled data.

Given a set of labeled data X_L, deep NN learn to generate the estimation function f through minimizing a cost function $L_{supervised}$, as follows:

$$\min_{\theta} \sum_{x \in X_L, y \in Y_L} L_{supervised}(x, y, \theta), \tag{1}$$

where θ denotes the model parameters. Due to the simplicity of integrating additional cost terms to the objective function of DNNs, a plethora of novel deep semi-supervised techniques have emerged, such as Ladder Networks [118], Deep Co-training [114], and graph NN [159]. Extending a supervised, NN-based learner to a semi-supervised learner can be performed by simply introducing an additional term denoted $L_{unsupervised}$ to the loss function as follows:

$$\min_{\theta} \left(\sum_{x \in X_L, y \in Y_L} L_{supervised}(x, y, \theta) + \sum_{x' \in X'_U} L_{unsupervised}(x', \theta) \right), \tag{2}$$

where $L_{unsupervised}$ represents an unsupervised loss computed on the unlabeled samples in the dataset. It is important to note that a variety of different loss functions and unsupervised loss terms can be employed and that different combinations of these terms yield distinct SSL models. A review of these models has been most recently published by Yang et al. in 2021 [151]. The authors survey major SSL techniques for image classification, which all share a deep NN backbone. Many of the surveyed methods are widely used in the context of semi-supervised SED, as the sound

Table 3. Summary of Semi-Supervised SED Methods and the Corresponding Loss Functions

Methods	Formulae
Mean Teacher [31, 32, 60, 97, 158], FixMatch [67], Random Consistency Training [123], Self-Training [39], Noisy Student [37], Unsupervised Data Augmentation [37], Cross Referencing Self-Training [108], Guided Multi-branch Learning [56], Hodgepodge [126], Guided Learning [82], Interpolation Consistency Training [66]	$BCE = -[y \log(p) + (1-y)\log(1-p)]$
Mean Teacher [31, 32, 35, 60, 97, 158], FixMatch [67], Random Consistency Training [123], Noisy Student [37], Unsupervised Data Augmentation [37], Cross-Referencing Self-Training [108], Hodgepodge [126], Interpolation Consistency Training [146]	$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$
Interpolation Consistency Training [146]	$HUBER = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{1}{2}z^2 \text{ if } \lvert z\rvert \leq \delta, \text{else } \delta\lvert z\rvert - \frac{1}{2}\delta^2\right)$
Mean Teacher [79], Pseudo-labeling [79]	$IOU = 1 - \frac{\lvert y \cap \hat{y}\rvert}{\lvert y\rvert + \lvert \hat{y}\rvert - \lvert y \cap \hat{y}\rvert}$
VAT [51], Self-Adaptive Label Refinement [51], Mean Teacher [35]	$KL(P\|\|Q) = \sum_i P(i)\log\left(\frac{P(i)}{Q(i)}\right)$
Mean Teacher [35]	$JSD(P\|\|Q) = \frac{1}{2}\left[KL(P\|\|M) + KL(Q\|\|M)\right]$

BCE, binary cross-entropy; IOU, intersection over union; JSD, Jensen-Shannon divergence; KL, Kullback-Leibler.

analysis community draws inspiration from image processing techniques. We refer the reader to an exhaustive list of SSL-based SED works and the corresponding choice of loss functions in Table 3.

6 Common Characteristics in Teacher-Student Techniques

Teacher-Student techniques typically involve training two models to predict sound events. The teacher model is a deep network trained on a large dataset of labeled audio signals, and the student model is trained to mimic the outputs of the teacher model on the unlabeled dataset [151]. Despite the differences in the specific implementations of these techniques, there are several common characteristics that underpin their effectiveness. In the remainder of this section, we will elaborate on the grouping of Teacher-Student methods represented in Figure 2. Specifically, we will discuss three common characteristics: ***Consistency Regularization (CR), Learner Multiplicity (LM),* and *Knowledge Distillation (KD)***. The diagram in Figure 2 shows that these characteristics are not mutually exclusive but rather overlap and interact with each other. Each of these characteristics contributes to the overall effectiveness of Teacher-Student techniques. Therefore, examining these shared characteristics will allow us to gain a deeper understanding of the underlying principles that govern these techniques and to provide insights that can guide future research in this area.

6.1 LM

For the past decade, deep learning techniques have become a cornerstone of machine learning for many sound analysis tasks such as sound scene classification [2], audio tagging [112], and SED [94]. In a conventional deep learning setting, a single learning model is fine-tuned through multiple iterations of training and is next deployed to generate inferences on unseen data [143]. In an attempt to further improve the performance of these learners, many previous works have focused

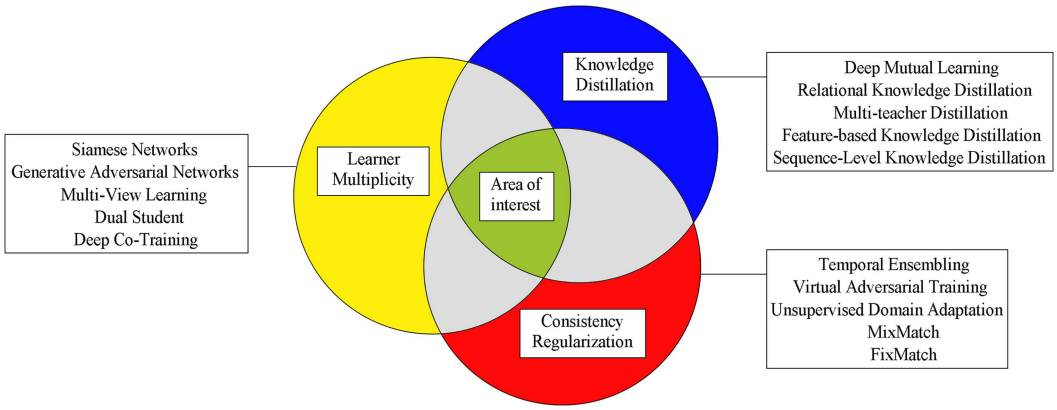


Fig. 2. Venn diagram illustrating the three key characteristics of Teacher-Student techniques. The overlapping circles represent the intersection of these characteristics, namely, KD, LM, and CR. In the center is the area of interest (green), which represents our Teacher-Student framework. The area of interest includes three Teacher-Student techniques: Mean Teacher, Interpolation Consistency Training, and Guided Learning.

on modifying the architecture design. The most notable alteration has been transforming these networks depth-wise and width-wise [134]. However, the ever-increasing number of parameters often renders these architectures challenging to train and deploy for two main reasons. First, a learner with millions of parameters requires a substantial amount of training resources and is unequivocally prone to high-latency training and prediction. Second, a complex learning algorithm often requires an exhaustive amount of training input to avoid the risk of overfitting. To overcome these hurdles, the interest has been directed toward reducing the footprint of a computational model by using multiple *compact* learners instead of one *cumbersome* learner.

One approach that illustrates the concept of LM is known in literature as ensemble learning [38]. Different from conventional deep learning pipelines, ensemble learning algorithms construct a set of approximation functions instead of a single function. The constructed functions are then aggregated in a specific fashion to produce the final output. Thus, ensemble-based systems are not constrained to a single learner but rather involve a *committee* of base learners. The ensemble method of learning has gained a great amount of popularity and has been adopted in a broad spectrum of works on audio analysis [14, 99, 100]. The superiority of ensembles over single learners resides in the combination of the predictions of multiple learners through either bagging, boosting, or averaging [38]. Ensembles can reduce the risk of overfitting when the base classifiers are trained on different subsets of the data. Ensembles can also be more robust than a single classifier, since the impact of individual errors or outliers can be reduced through the combination of predictions [38]. The effectiveness of ensemble stratagems has contributed to the emergence of a large body of literature on multiple learner-based systems, we refer the reader to a comprehensive list of training methods that encompass multiple learners in Table 4.

Another illustration of this concept are KD pipelines. In distillation-based procedures, knowledge about a given task is distilled from a large computational model or an ensemble of models to a lighter footprint model [49]. In these settings, it is imperative to include more than a single network into the training process. Bucilua et al. [15] proposed a method for transferring knowledge from large complex ensembles to small models. The devised method allows compact NN to learn a function by mimicking a very large ensemble of base-level classifiers. This is achieved via using the ensemble to label large unlabeled datasets, which are later used to train a compact NN. In this training setup, the ensemble acts as a teacher or rather an ensemble of teachers, while the target

Table 4. Comprehensive List of Deep Learning-Based Training Methods with Multiple Learners

Methods	Learners	Identical Learners	Applications
Mean Teacher [136]	2	Yes	Semi-Supervised Object Detection [151], Cross-Domain Object Detection [33], Brain Lesion Segmentation [29], Sound Event Separation and Classification [124]
Tri-training [161]	3	Yes	Unsupervised Domain Adaptation [119], Web page classification [161], Recommender Systems [152]
Ensemble Methods [38]	Varies	No	Medical Image Classification [83], Sequence Classification [150], Electric Load Forecasting [115], Computer Vision [24], Natural Language Processing [83]
Deep Ensemble Methods [156]	Varies	No	Brain Disease Diagnosis [133], Alzheimer's Disease Classification [7], Monaural Speech Separation [156]
KD [120]	2	No	Object Detection [23], Anomaly Detection [120], Natural Language Processing [50]
Multi-Task Learning [117]	2 or more	No	Text Classification [84], Natural Language Processing [27], Speech Recognition [34], Computer Vision [44], Drug Discovery [117]
Adversarial Multi-Task Learning [144]	2 or more	No	Robust Speech Recognition [127], Emotion Classification [80], Face Sketch Synthesis and Recognition [144]
Co-training [80]	2 or more	Yes	Medical Image Segmentation [148], Automatic Image Annotation [80], Fault Detection and Classification [1], Well Condition Recognition [159]
Federated Learning [129]	2 or more	No	MRI Analysis [116], Human Activity Recognition [129], Detect Credit Card Fraud [151]
Generative Adversarial Networks [55]	2	No	Texture Synthesis [78], Face Synthesis [55], Image Super-Resolution [77], Sentence Generation [81], Anomaly Detection in Medical Imaging [121]
Reinforcement Learning with Multiple Agents [10]	2 or more	No	Control of Traffic Signals [10], Navigation [13], Area Sweeping [88], Automated Trading [132], Network Routing [137]

model acts as the student. The experimental results show that small footprint models are 1,000 times faster than their counterpart large ensembles.

Parallel to the work of Bucilua et al. [15] is Teacher-Student learning as introduced by Hinton et al. [54]. In their work, the authors show that it is feasible to transfer knowledge from a heavy, highly regularized teacher model to a small distilled student. The distillation process employs soft targets generated by the cumbersome teacher network or teacher ensemble. The student model is later trained to match the soft targets produced by the teacher network. In the same line of

work, Mirzadeh et al. introduced a novel training scheme, which involves more than a single network, termed **Teacher Assistant KD (TAKD)** [95]. In their work, Mirzadeh et al. assert that the architectural gap between the student and the teacher network can result in a degradation of the student performance. To mend this, they propose a multiple learner-based pipeline in which an intermediate-size network is employed to bridge the gap between the teacher and the student. The authors demonstrate the effectiveness of using one or multiple teacher assistants through rigorous empirical and theoretical analysis.

Multiple learner-based strategies have also been extended to learning schemes which dispose of the teacher in the training loop. In deep mutual learning [157], it is argued that no complex teacher network is necessary to teach a lightweight student. In their work, Zhang et al. propose a training strategy where an ensemble of student networks collaborate to teach each other [157]. The experimental results reveal that mutual learning between simple compact students outperforms the typical distillation from a powerful teacher strategy.

6.2 CR

Deep learning architectures allow for learning representations of the raw data with multiple levels of abstraction and are highly data-driven methods, i.e., they require large volumes of data for optimal generalization performance [143]. In consequence, deep architectures are extremely prone to overfitting, which makes regularization an essential step of a deep learning pipeline [71]. Kukačka et al. categorized regularization techniques based on where it occurs during NN training [71]. According to their work, regularization methods can be categorized into five major families: regularization via data, via the network architecture, via the error function, via the regularization term, and via optimization. Regularization via the regularization term is achieved by adding a penalty term to the error function of model. A regularization term is independent of the target, thus making it highly suitable when unlabeled samples are available. The value of a regularization term can thus be computed for an unlabeled point and incorporated with the error function computed on labeled points. This formulates the basis of CR methods in SSL.

CR methods have demonstrated their ability to yield notable results in deep SSL [151]. It leverages the use of abundant unlabeled data based on the following intuition: *the prediction of a learning model should not change under any realistic perturbation of the data or the model* [151]. It is worth mentioning that the proper choice of the perturbation method is critical to the effectiveness of CR-based techniques [104]. Concretely, consistency between model predictions is achieved by introducing an auxiliary loss term, also referred to as a consistency term, to the final loss function of the model. Minimizing the consistency cost corresponds to minimizing the distance between the two outputs of the model on the perturbed and non-perturbed version of the data point. The most commonly used distance measures to quantify consistency are MSE, **Jensen-Shannon divergence (JSD)**, and **Kullback-Leibler (KL)** divergence [104]. Details about the formulas used to compute the aforementioned measures can be found in Table 3.

The concept of CR can be found in a number of previous works on SSL. One notable mention is the work of Samuli Laine and Timo Aila where they propose Temporal Ensembling [74]. In their work, the authors train a model on both labeled and unlabeled samples while keeping a memory buffer to store the predictions on the unlabeled points. The buffer is updated with the predictions made on the current batch of unlabeled data during each iteration. The stored predictions are used to compute a target distribution and minimize the distance between the predicted and target distributions, thus enforcing CR.

Inspired by Adversarial Training [46], Miyato et al. propose **Virtual Adversarial Training (VAT)** [96], a CR-based method that enforces the predictions of a model to remain stable under small perturbations in the adversarial direction of the input. To accomplish this, VAT utilizes an

adversarial perturbation generated from the current input. The model is trained to predict similar outputs for the original and perturbed inputs, forcing the model to be smooth around each input data point. The term "virtual" in VAT comes from the fact that the perturbation is approximated without any labels, making it applicable to SSL.

In DCASE2016, Matthias Zohrer and Franz Pernkopf used VAT to regularize **Gated Recurrent NN (GRNNs)** and improve the performance of sound event detectors [166]. The authors conducted experiments on two different datasets and achieved state-of-the-art results compared to previous methods. Similarly, Anthony Agnone and Umair Altaf proposed a sound detection system based on VAT to improve the generalization abilities of their detectors [6]. The authors also employ data augmentation techniques, such as altering the pitch and speed of the audio signals or random time-shifting to increase the robustness of the model. The system was evaluated on the DCASE2019 Task 4 dataset and achieved competitive results, demonstrating the effectiveness of using VAT and CR in AED.

MixMatch is another method that involves CR for SSL [12]. Berthelot et al. proposed a holistic method that combines both CR and entropy minimization into a single loss [12]. The particularity of this method is the use of Mixup regularization to combine pairs of augmented labeled and unlabeled samples. MixMatch allows the model to learn from both labeled and unlabeled data simultaneously, which can lead to better performance, especially when labeled data is limited.

Inspired by the triumph of MixMatch, Shi et al. adapted the latter technique to a multi-hot classification setting, thus introducing Multi-hot MixMatch (M3) [125]. The proposed method was evaluated on the DCASE2018 task 4 dataset, achieving state-of-the-art results on both the development and evaluation sets. The results further show the effectiveness of CR in detecting rare sound events.

6.3 KD

Scaling deep learning architectures allows for designing highly complex and elaborate learning models that can maneuver an extensive number of parameters. Nevertheless, a greater computational complexity comes at a cost during the deployment phase. For instance, SED systems aimed for acoustic surveillance must be deployable on equipment with limited computational power and must operate in real time [143]. However, complex deep learning systems are usually computationally exhaustive and are thus not efficient in resource-constraint scenarios. To overcome these hurdles, an interesting line of research work has been devoted to what is known as model compression [25]. The latter enables the reduction in computational complexity of a heavy model, without substantially impacting the performance. Several approaches have been proposed such as parameter pruning [162] and network quantization [45]. A summary of notable works on model compression and acceleration has been provided by Cheng et al. in 2018 [25].

KD-based methods are one of many categories of methods for model compression according to the aforementioned survey [25]. It is the process by which the knowledge from one model, termed as a teacher, is transferred or *distilled* to another model, termed as a student [25]. KD has recently received renewed interest as a methodology for knowledge transfer. The intuition behind transferring learned information from a teacher to a student is motivated by the notion of *dark knowledge* [54]. As pioneered by Hinton et al., dark knowledge represents the "hidden" information captured by the teacher model during training. One popular and effective approach for distilling this knowledge is to train a student model to match the soft targets produced by a teacher model [54]. Soft targets, as defined by Hinton et al., are the output probabilities generated by the teacher model. The intuition behind this approach is as follows: soft targets encapsulate much richer information about a given task than hard targets. For instance, given an audio excerpt, a network trained for SED assigns different probability values for each occurring event.

This in term better matches the uncertainty encountered in a real environment, for example, a person laughing and a person crying both produce similar sounds in terms of pitch and volume, especially when the laughter is hysterical or the crying is intense. Therefore, if the aim is to construct a better student model, it should learn to mimic the behavior of the teacher instead of hard targets.

KD has been recently extended to the SSL paradigm [103]. It is assumed that massive quantities of available unlabeled data can carry a considerable number of informative cues about the data distribution. Therefore, extracting and distilling the knowledge acquired via learning from these cues can relieve the detrimental effect caused by the absence of strongly labeled data [151]. Papernot et al. considered KD as a means of transferring knowledge in a scenario where conserving data privacy is of high priority [107]. The authors devised a black-box approach coined "Private Aggregation of Teacher Ensembles" that combines numerous models trained on disjoint subsets of sensitive data. The aforementioned models are used as teachers for a student model that learns to match an output chosen by a noisy voting strategy among all the teachers.

Most recently, Li et al. [79] published a research work on learning from weakly labeled web video content, where they proposed to convert the potential noise present in weak labels as additional supervision signals. The authors introduced the concept of **Sub-Pseudo Label (SPL)** and argued that it spans out a different set of meaningful "middle ground" label space to pre-train a teacher network. They show that the combination of SPL and the distilled knowledge from the teacher network helps enrich the supervision signals of the student in consequent training stages.

Shi et al. investigated the performance of a Teacher-Student training approach on the large-scale weakly labeled Audioset dataset [126]. They initially showed that state-of-the-art performance can be achieved by introducing an attention mechanism to a convolutional NN model. Furthermore, they devise an end-to-end Teacher-Student framework to distill knowledge from a large SED teacher model to a small-footprint student, while maintaining good detection performance. The experimental results suggest that distilling knowledge from attention-equipped teacher models can help design robust student models with high SED rate.

Most recently, a paper published by Anurag Kumar and Vamsi Krishna Ithapu proposed a sequential self-teaching approach for sound event recognition [73]. The inspiration behind the author's proposition is the sequential nature of human learning over time. The proposed framework involves training a sequence of NN on weakly labeled audio data. At each stage, a network from the sequence of trained networks is considered a student and is guided by the network(s) from the previous stage(s). Concretely, knowledge is distilled throughout the training stages in the form of "co-supervision," i.e., the student in the current stage learns from the ground truth labels and the outputs of one or more teachers from previous stages, and consequently becomes a teacher in the next stage.

Koutini et al. proposed an iterative KD technique for weakly labeled semi-supervised SED [70]. Specifically, they employed a CRNN to predict pseudo-labels, which are then used for KD in an iterative fashion. They showed that the iterative nature of this process allows for more accurate onset and offset detection of sound events after each iteration.

It is worth indicating that transferring knowledge from a teacher to a student is the basic principle echoed throughout KD literature. Nonetheless, the strategies developed for the aforementioned purpose often differ from each other depending on multiple criteria. We refer the reader to a comprehensive survey on KD and Teacher-Student learning, published by Lin Wang and Kuk-Jin Yoon in 2021, in which they address the most fundamental questions driving the research on the topic [145].

7 Teacher-Student Framework for SED

7.1 Mean Teacher

Mean Teacher is an SSL approach that has gained increasing attention in recent years, due to its effectiveness in achieving state-of-the-art performance on various tasks [136]. Mean Teacher was first introduced for image classification by Tarvainen and Valpola in 2017. The approach builds upon the foundations of Temporal Ensembling and is further improved to overcome the limitations thereof [74]. Temporal Ensembling works by aggregating the predictions of the model over time, which ultimately forms an ensemble prediction that the algorithm uses as a consistency target. This helps to reduce the impact of noisy predictions and provides a more stable estimate of the underlying structure of the unlabeled space. Nonetheless, Temporal Ensembling suffers from a major drawback: the learned information is incorporated into the training process at a slow pace since each target is updated only once per epoch. Travainen and Valpola proposed their algorithm to circumvent the latter drawback by aggregating model weights instead of model predictions [136].

Mean Teacher *explicitly* satisfies the concept of LM in the Teacher-Student framework by training a pair of networks: a teacher and a student. The two networks are trained simultaneously on batches of labeled and unlabeled data and the student is trained to minimize a composite loss function as follows:

$$L_{MT} = L_{supervised} + \lambda \cdot L_{unsupervised} \tag{3}$$

where λ controls the relative weight assigned to the unsupervised loss, i.e., consistency loss during training. The supervised loss term $L_{supervised}$ is computed using the predictions of the student on the labeled batch of the training data X_L, while the unsupervised term $L_{unsupervised}$ is computed as the distance between the outputs of the teacher and student model on unlabeled data as shown in Algorithm 1. One critical specification in Mean Teacher resides in the architecture of the two networks: both the student and the teacher models must be *identical*. This is to ensure that the update of the teacher weights can be properly executed using the **Exponential Moving Average (EMA)** rule. EMA smoothing is a statistical technique that computes a weighted average of a time series data, it assigns more weight to recent observations and gradually decreases the weight of older observations, resulting in a smoothed series as shown in Figure 3. At each training step t, the student weights θ_t are updated using a standard backpropagation algorithm, while the teacher weights θ'_t are computed using the EMA rule as follows:

$$\theta'_t = \alpha \cdot \theta'_{t-1} + (1 - \alpha) \cdot \theta_t, \tag{4}$$

where θ'_{t-1} represents the weight of the teacher model from the previous time step. The α term is a crucial hyperparameter known as the smoothing coefficient. This coefficient and its complement are used to regulate how far the EMA reaches into the training history and to control the influence of both networks into the final weight update.

It is worthy of note that Mean Teacher also *implicitly* satisfies the concept of LM through the aggregation of multiple teachers at every training step. The weights of the teacher network are averaged over time to generate an ensemble of teacher models, and the student model is trained to mimic the ensemble [136]. Along the same lines, the teacher network generates soft labels (i.e., probability distributions over the output space) for the unlabeled data, which serve as targets for the student network. The student attempts to match these soft targets throughout the training process, in other words, the student learns to mimic the teacher thereby satisfying the KD axe of our Teacher-Student framework.

Another specification for the Mean Teacher algorithm is the injection of noise in the training process. In their work, Tarvainen and Valpola evaluated their algorithm using both input perturbations, such as random translations and horizontal flips, and perturbations in the intermediate

Algorithm 1 Pseudocode of Mean Teacher

Require: Labeled dataset D_L, Unlabeled dataset D_U, Teacher network T with parameters θ', Student network S with parameters θ, Consistency Coefficient λ
Ensure: A trained teacher network with parameters θ'
  for each epoch **do**
    for each iteration t in minibatch B **do**
      Apply noise η on each minibatch in B
      Compute the outputs with respect to noise η using the student network by performing a forward pass on B
      Apply noise η' on each minibatch in B
      Compute the outputs with respect to noise η' using the teacher network by performing a forward pass on B
      Compute the supervised classification loss of the student network:
        $L_{\text{supervised}} = \text{BCE}(S(\eta'(x_i), \theta'), y_i)$
      Compute the unsupervised consistency loss between the teacher and student logits:
        $L_{\text{unsupervised}} = \text{MSE}(T(\eta(x_i), \theta), S(\eta'(x_i), \theta'))$
      Compute the overall loss by weighing the unsupervised loss using a consistency cost λ:
        $L_{\text{MT}} = \text{BCE}(T(\eta(x_i), \theta), y_i) + \text{BCE}(S(\eta'(x_i), \theta'), y_i) + \lambda \times \text{MSE}(T(\eta(x_i), \theta), S(\eta'(x_i), \theta'))$
      Update the teacher network weights using the exponential moving average update rule:
        $\theta'_t = \text{EMA}(\theta'_{t-1}, \theta_t, \alpha)$

representations through dropout and Gaussian noise [136]. The intuition behind noise injection is to introduce variability into the unlabeled data and encourage the model to learn more robust and generalizable representations. Mean Teacher thus satisfies the axe of CR in the Teacher-Student framework in two ways. First, the student model is encouraged to produce consistent predictions on the labeled portion of the data with respect to input perturbations. Second, the student is encouraged to produce consistent predictions on unlabeled data with respect to teacher's predictions.

Mean Teacher-based strategies have been adopted for sound analysis tasks and have recently been exhaustively used for semi-supervised SED. In DCASE2018, Lu JiaKai proposed a Mean Teacher-based system for large-scale detection of sound events using weakly labeled data [60]. The system was built using a stacked context-gating convolutional NN, a recurrent NN, and an attention module. Lu JiaKai's detector was successful at exploiting large amounts of unlabeled and unbalanced training data and achieved the highest scores during evaluation thereby winning the competition. Owing to the triumph of the aforementioned system, the baseline system for DCASE2019 was based on the same idea [111]. Consequently, the overwhelming majority of the systems submitted for evaluation in DCASE2019 were based on the Mean Teacher approach. Arguably, the most interesting aspect of the DCASE2019 outcomes are the compelling results of Mean Teacher-based detectors such that the top-performing submissions used the latter approach.

Delphin-Poulat and Plapous proposed a Mean Teacher-based sound event detector inspired by the baseline and mainly focused on applying data augmentation [32]. A total of six other submissions based on the Mean Teacher approach have succeeded at overthrowing the provided baseline, which demonstrates the effectiveness of this algorithm for PSED under the SSL paradigm [111].

The favorable performance of this approach across numerous deep learning tasks has inspired a plethora of Mean Teacher variations up-to-date. In Humble Teacher, Tang et al. proposed a novel semi-supervised technique for object detection that achieved state-of-the-art performance on multiple benchmarks [135]. Furthermore, Tang et al. demonstrated the efficiency of the EMA iteration-wise updates of the teacher model. The authors compare an EMA-updated teacher against

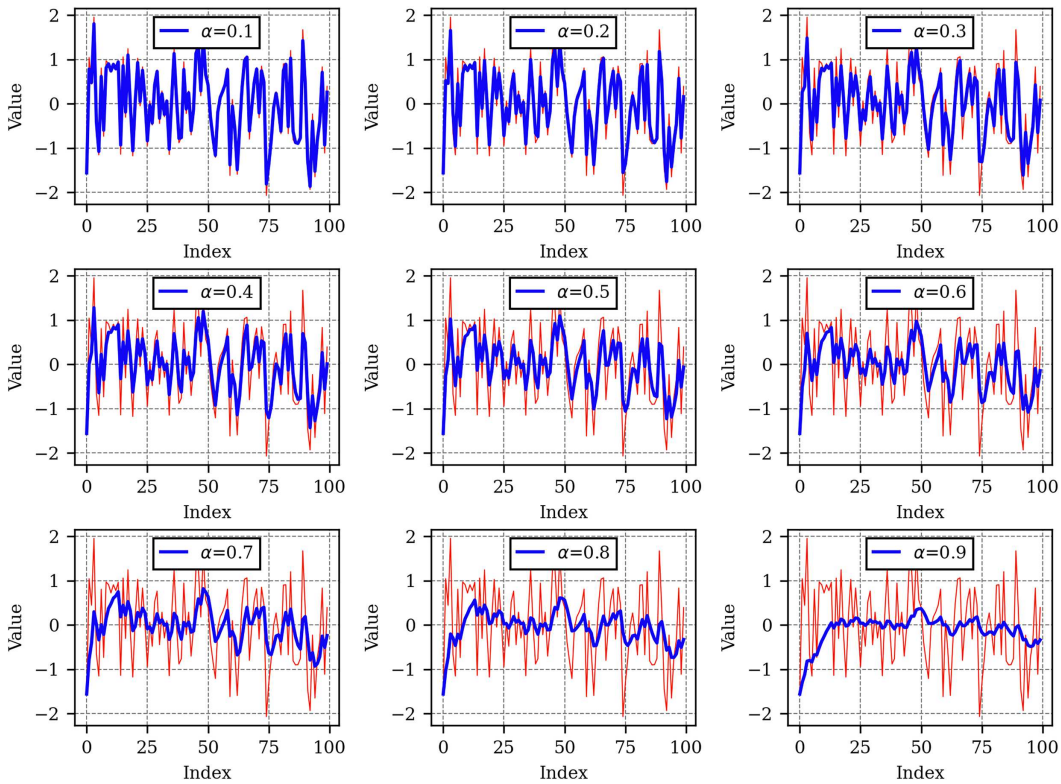


Fig. 3. EMA update rule for different values of the smoothing parameter α. Each subplot shows the original data (red) and the EMA-smoothed data (blue) with the corresponding α value. The figure highlights how increasing the α value results in a smoother EMA curve that more heavily weights recent data.

a teacher for which the weights have been copied every 10K iteration and concluded that the detrimental effect of incorrect pseudo-labels is mitigated by EMA. Soft Teacher is another semi-supervised object detection approach that successfully adapts a Mean Teacher algorithm for object detection [149]. In their work, Xu et al. proposed a framework that discards the complex multi-stage methods that previously exist [149]. The teacher model is continuously updated by the student model using the EMA strategy for online pseudo-labeling.

7.2 Guided Learning

Guided Learning is an SSL approach proposed by Liwei et al. in 2019 [82]. The method was specifically developed for the SED task during DCASE2019. The Guided Learning-based system was able to yield state-of-the-art sound event detectors, thereby winning the challenge. In their work, Liwei et al. primarily focused on avoiding the customary tradeoff between the two subtasks involved in PSED, namely, audio tagging and boundary detection. The authors argue that a large compression scale enables an NN to produce more accurate audio tags but less accurate event boundaries [82] and that previous work commonly seeks this tradeoff by choosing a moderate temporal compression scale. Liwei et al. proposed a method which employs a pair of models to pursue the two subtasks independently. Specifically, the approach aims to avoid the tradeoff and ensure favorable performance for both subtasks.

The Guided Learning approach *explicitly* satisfies the concept of LM by employing two deep NN. One network is denoted as a professional teacher, while the other network is denoted as a

promising student. The primary distinction between Guided Learning and other Teacher-Student techniques lies in the design choice of the networks. In Guided Learning, the student model is intentionally designed to be inferior to the teacher network to ensure that the teacher *guides* the student. Furthermore, this configuration directly exemplifies the KD concept of our Teacher-Student taxonomy, with a single exception: the aim is not to reduce the model size but rather to design a student model that is competitive or even superior to the teacher model at a given task.

To guarantee the architectural superiority of the teacher network, Liwei et al. designed the student network to be shallower and endowed the teacher network with a larger sampling scale [82]. This ensures that the network possesses a large receptive field, which allows better exploitation of contextual information, thus enabling the teacher to be highly performant at audio tagging. Alternatively, the student network is equipped with a smaller sampling scale. A narrow receptive field enables the student network to detect fine-level information, thus allowing high performance in temporal boundary detection [85]. One important characteristic of the Guided Learning method resides in the manner in which knowledge is distilled. In most KD schemes, the transfer of knowledge is unidirectional, which is usually distilled from a deeper/complex network to a shallower/compact network [145]. Conversely, knowledge in Guided Learning is distilled in a bidirectional fashion, both models help fine-tune each other throughout the training process, which raises their abilities in audio tagging and event detection alternately. To achieve this, Guided Learning minimizes a composite loss $L_{GuidedLearning2}$ as shown in Algorithm 2. However, the latter loss is only updated during the last s epochs of the setup. During the early epochs, an initial loss is minimized as follows:

$$L_{GuidedLearning1} = L_{PS} + L_{PT} + L_{unsupervised}, \tag{5}$$

where both the teacher and student models learn from the provided ground truth via the supervised loss $L_{supervised}$. Simultaneously, only the student is updated through an unsupervised loss as follows:

$$L_{supervised} = L_{PS} + L_{PT} \tag{6}$$

$$L_{unsupervised} = J(\phi(T(x, \theta')), S(x, \theta)), \tag{7}$$

where $\phi(\bullet)$ computes the clip-level output of the teacher model as shown in Algorithm 2. The $L_{unsupervised}$ component helps the student model learn from a pool of unlabeled data with audio tags generated by the teacher model. The latter is guaranteed by the superiority of the professional teacher.

It is worthy of note that at this stage, Guided Learning enforces the promising student to be consistent with the audio tagging predictions of the professional teacher, which satisfies the CR axe of our taxonomy. The authors argue that as the training progresses, the faculty of the student model gradually approaches that of the teacher model. Therefore, the teacher model additionally keeps an unsupervised loss $L'_{unsupervised}$ as follows:

$$L'_{unsupervised} = J(\phi(S(x, \theta)), T(x, \theta')). \tag{8}$$

At this stage, Guided Learning enforces the professional teacher to be consistent with the audio tagging predictions of the promising student. Ultimately, during the last training epochs, Guided Learning minimizes a different loss where knowledge is distilled from the teacher to the student and vice versa as follows:

$$L_{GuidedLearning2} = L_{supervised} + L_{unsupervised} + a \cdot L'_{unsupervised}, \tag{9}$$

where a is a factor which increases gradually throughout the training process as indicated in Algorithm 2. Liwei et al. showed that the Guided Learning approach outperforms state-of-the-art Mean Teacher-based systems and demonstrated the potential of Teacher-Student methods for PSED. Furthermore, due to the disparity in architecture between the teacher and student networks, the

Algorithm 2 Pseudocode of Guided Learning

Require: Labeled dataset D_L, Unlabeled dataset D_U, Professional Teacher PT with parameters θ', Promising Student PS with parameters θ, Increasing factor $a \leftarrow 0$, Start epoch of the second stage s, Hyperparameter γ
Ensure: A trained student network with parameters θ
  for each epoch **do**
    if epoch > s **then**
      $a = 1 - \gamma^{(epoch-s)}$
    for each iteration in minibatch B **do**
      Compute the supervised loss of the student network by performing a forward pass on $B \cap D_L$:
        $L_{PS} = \text{BCE}(y_i, PS(x_i, \theta))$
      Compute the supervised loss of the teacher network by performing a forward pass on $B \cap D_L$:
        $L_{PT} = \text{BCE}(y_i, PT(x_i, \theta'))$
      Compute the clip-level predictions of the student and the teacher networks with respect to $B \cap D_U$:
        $\phi(PS(x_i, \theta)) = \begin{cases} 1, & \text{if } PS(x_i, \theta) > 0.5 \\ 0, & \text{otherwise} \end{cases}, \quad \phi(PT(x_i, \theta')) = \begin{cases} 1, & \text{if } PT(x_i, \theta') > 0.5 \\ 0, & \text{otherwise} \end{cases}$
      Compute the unsupervised loss of the student network by performing a forward pass on $B \cap D_U$:
        $L_{\text{unsupervised}} = \text{BCE}(\phi(PT(x_i, \theta')), PS(x_i, \theta))$
      Compute the unsupervised loss of the teacher network by performing a forward pass on $B \cap D_U$:
        $L'_{\text{unsupervised}} = \text{BCE}(\phi(PS(x_i, \theta)), PT(x_i, \theta'))$
      Compute the total loss by combining all 4 losses:
        $L_{GL} = L_{PS} + L_{PT} + L_{\text{unsupervised}} + a \times L'_{\text{unsupervised}}$
      Update θ and θ' neural network parameters using the computed gradients of the total loss

Guided Learning approach offers the potential of employing state-of-the-art models that have previously been established for each individual subtask [82].

7.3 Interpolation Consistency Training

Interpolation Consistency Training is an SSL method proposed by Verma et al. in 2019 and has recently gained attention for its effectiveness and computational efficiency [141]. Verma et al. originally designed and tested the Interpolation Consistency Training method on image classification benchmarks and showed that it can improve the performance of standard deep network architectures through data-adaptive regularization. The method was proposed to overcome the limitations of a previously established training technique known as VAT [96]. VAT is an SSL method proposed by Miyato et al. that has shown to be highly effective [96]. The adversarial examples generated using VAT effectively encourage the model to learn features that are robust to small perturbations. One of the primary limitations of VAT is the additional computational overhead associated with generating adversarial examples during the training process, which can increase the overall training time and resource requirements. Verma et al. later proposed Interpolation Consistency Training to overcome these limitations [141].

Algorithm 3 Pseudocode of Interpolation Consistency Training

Require: Labeled dataset D_L, Unlabeled dataset D_U, Teacher network T with parameters θ', Student network S with parameters θ, Random distribution β on $[0, 1]$
Ensure: Trained teacher network with parameters θ'
  for each epoch **do**
    for each iteration t in minibatch B **do**
      Update the student network by performing a forward pass on $B \cap D_L$ and computing the output logits
      Compute the supervised loss, and backpropagate through the student network:
        $L_{\text{supervised}} = \text{BCE}(S(x_i, \theta), y_i)$
      Compute pseudo labels for two randomly drawn samples in $B \cap D_U$ using the teacher network:
        $\hat{y}_i = T(x_i, \theta')$, $\hat{y}_j = T(x_j, \theta')$
      Sample interpolation coefficient λ from $\lambda \sim \beta$
      Compute the interpolation of the two randomly drawn samples in $B \cap D_U$ using Mixup:
        $x_{\text{mix}} = \lambda \cdot x_i + (1 - \lambda) \cdot x_j$, $y_{\text{mix}} = \lambda \cdot \hat{y}_i + (1 - \lambda) \cdot \hat{y}_j$
      Compute the unsupervised loss i.e. consistency loss term:
        $L_{\text{unsupervised}} = \text{MSE}(S(x_{\text{mix}}, \theta), y_{\text{mix}})$
      Compute the final loss and compute the gradients:
        $L_{\text{ICT}} = L_{\text{supervised}} + \omega(t) \cdot L_{\text{unsupervised}}$
      Update the teacher network weights using the exponential moving average update rule:
        $\theta'_t = \text{EMA}(\theta'_{t-1}, \theta_t, \alpha)$

Interpolation Consistency Training is built following the Mean Teacher structure, where the parameters of the teacher network θ' are an EMA of the student network parameters θ as shown in Equation (4) [136]. Hence, Interpolation Consistency Training *explicitly* satisfies the LM axe of our Teacher-Student taxonomy. Furthermore, since the backbone of Interpolation Consistency Training is a Mean Teacher, the aim of the student network is to match the targets generated by the teacher network, thereby satisfying the KD axe of our Teacher-Student framework. In Interpolation Consistency Training, both networks have access to labeled and unlabeled samples, and the ultimate learning goal is to minimize a composite loss function at each iteration as follows:

$$L_{\textit{Interpolation Consistency Training}} = L_{supervised} + \omega(t) \cdot L_{unsupervised}, \tag{10}$$

where the first component of $L_{\textit{Interpolation Consistency Training}}$ is computed on the labeled subset of each minibatch as shown in Algorithm 3. The second component of $L_{\textit{Interpolation Consistency Training}}$ is computed on the unlabeled samples of each minibatch and is weighted using a ramp function $\omega(t)$ to regulate the importance of the unsupervised loss after each training iteration.

The Interpolation Consistency Training method *explicitly* employs the concept of CR through the unsupervised loss term which acts as a regularization term and encourages the student network to be consistent with the outputs of the teacher network. Specifically, the student network is regularized to encourage the prediction at an interpolation of unlabeled data points to be consistent with the interpolation of the teacher predictions at these data points, which sets the Interpolation Consistency Training method apart from other Teacher-Student methods.

Formally, given two data samples x_1 and x_2 drawn from the unlabeled distribution D_U, Interpolation Consistency Training implements the concept of CR as follows:

$$S(I(x_1, x_2), \theta) \approx I(T(x_1, \theta'), T(x_2, \theta')), \tag{11}$$

where I denotes the interpolation policy applied on the two unlabeled points. In their work, Verma et al. [141] opted for a computation efficient function known as Mixup [154]. Proposed by Zhang et al. in 2018, Mixup has proven to be an effective regularization method in a diverse range of learning tasks including sound analysis tasks [36]. Mixup acts as a regularizer through generating feature-target instances from the vicinity distribution of training examples, which helps push the decision boundary of a learner further from the class boundaries. In the presence of labeled samples, Mixup constructs new synthetic samples by convexly combining two randomly drawn data points from the input space, both in terms of the raw feature vectors and ground truth encodings. Concretely, the interpolated sample $(\hat{x}, \hat{y})$ can be defined as follows:

$$\hat{x} = \lambda \cdot x_i + (1 - \lambda) \cdot x_j \tag{12}$$

$$\hat{y} = \lambda \cdot y_i + (1 - \lambda) \cdot y_j, \tag{13}$$

where x_i and x_j are two randomly selected feature vectors and y_i and y_j are their corresponding true class labels, respectively. The linear interpolator λ is sampled from a Beta distribution [36]. The Beta probability distribution is continuous, symmetric, and is regularized by a hyperparameter known as α. The latter hyperparameter affects the degree of interpolation between the original examples in the synthetic example [76]. Prior work has proven the efficiency of the Mixup regularization method improving the generalization ability of deep NN [36]. Therefore, the method has recently been extended to the semi-supervised setting [12], by incorporating interpolations of unlabeled data points into the training process such as in Interpolation Consistency Training. Under a semi-supervised setting, the interpolation policy using Mixup is equivalent to Equations (12) and (13), except that in the absence of ground-truth labels, an estimated target or "fake label" is generated and substituted for ground truth. In Interpolation Consistency Training, these "fake labels" are generated using the teacher network function with parameters θ', hence, the formula for training a semi-supervised learner under the Interpolation Consistency Training regime becomes as follows:

$$S((\lambda \cdot x_1' + (1 - \lambda) \cdot x_2'), \theta) \approx \lambda \cdot T(x_1', \theta') + (1 - \lambda) \cdot T(x_2', \theta'). \tag{14}$$

Due to its simplicity and minimal computation overhead, the Interpolation Consistency Training method has been recently adopted in semi-supervised SED. In 2019, Ziqiang Shi proposed a semi-supervised technique known as Hodgepodge for SED [126]. The author adapted the Interpolation Consistency Training approach for sound data and yielded state-of-the-art results for semi-supervised PSED. The developed systems were based on a CRNN network architecture and were trained on interpolated samples drawn from the DESED dataset. Furthermore, Ziqiang Shi ensembled Interpolation Consistency Training-based models in an attempt to improve the detection performance. To achieve this, the author ensembled models based on the value of the Mean Teacher consistency coefficients and showed that ensembles using Hodgepodge are effective in detecting sound events in the absence of labeled data.

Inspired by Interpolation Consistency Training and Hodgepodge, Koh et al. proposed a semi-supervised method known as **Shift Consistency Training (SCT)** [68]. The approach encourages the prediction of time-shifted input vectors to be consistent with time-shifted predictions of the same input vector. The authors theorize that this training routine allows semi-supervised learners to learn shift-invariances in acoustic features which ultimately helps localize sound events in time. The method was tested on the DESED corpus and submitted for evaluation during DCASE2020. The authors showed the SCT method helps the model learn from ambiguous samples which boosts the performance of the detectors. The authors also introduce **Clip-level Consistency Training (CCT)** in which they enforce consistency between the clip-level outputs of an architecturally superior network and the frame-level outputs of a less complex network. The aim of this approach is to enhance the network's ability to extract the appropriate acoustic features.

8 Experimental Setup

8.1 Training Corpora

For the purpose of our study, we have opted for the TUT Sound Events 2017 dataset (TUT-SED2017) [90]. The dataset consists of recordings capturing street acoustic scenes with varying levels of traffic and other human activities. The dataset was manually annotated by a single annotator with all audible sound events, in addition to the start and end times of each event occurrence. The target classes were selected based on the frequency of obtained annotations, which include *brakes squeaking, car, children, large vehicle, people speaking*, and *people walking*.

The TUT-SED2017 dataset comprises two subsets: a development set and an evaluation set. A four-fold cross-validation setup was provided with the dataset, with each recording used exactly once as test data. However, during our experiments, we deviated from the provided cross-validation setup. After conducting preliminary tests, we found that the cross-validation setup did not align with our experimental requirements. The reason is that we progressively discarded large ratios of labels, resulting in a smaller development set. Therefore, we merged all four folds of the cross-validation setup and opted for a train-test-split approach, with 80% of the data used for training and 20% for testing. To obtain unlabeled data samples for our study, we followed a common practice in the literature known as artificial unlabeling [48]. This approach involves systematically removing the labels from a substantial portion of data points within an existing SL dataset. By adopting this technique, we simulate various data settings by considering different quantities of labeled samples to evaluate our SSL algorithms. Furthermore, we diligently maintained a fixed random seed value throughout our experimental implementation to ensure the reproducibility of our results. This allowed us to control the generation of random data splits and minimize potential variations, enabling accurate comparisons and reliable validation of our results.

8.2 Architectural Design

To build our polyphonic sound event detectors, we have adopted a CRNN architecture. Our choice of architecture is consistent with the prevailing trend in the field, as CRNNs have been widely and extensively employed by the research community. More specifically, for our Mean Teacher and Interpolation Consistency Training-based detectors, we employ a stack of three convolutional blocks, which are succeeded with a single recurrent block and a fully connected layer. The architecture of the convolutional block is detailed in Table 5. The RNN component is composed of two layers of **Bi-directional Gated Recurrent Units (Bi-GRU)**, the architecture of the RNN block is detailed in Table 5. Note that this design is identical for both the teacher and student components.

For our Guided Learning-based detector, the specifications for the convolutional block of the teacher component can be found in Table 5. The model comprises a stack of three convolutional blocks followed by three layers of Bi-GRU. For the student network, we maintain the same architectural design for the convolutional blocks as with Mean Teacher and Interpolation Consistency Training and only employ a single layer of Bi-GRU for the RNN block.

During all of our experiments, we train our systems for 50 epochs using Mel-spectrograms as our input features. To extract the input features, we apply a Short Time Fourier Transform to each audio clip using a Hamming window of 2,048 samples and a hop size of 511 samples. The resulting spectrogram is then processed through 40 Mel filter banks to obtain Mel-spectrogram feature matrices. We have implemented all of our detectors using Pytorch and optimized the parameters using the ADAM optimizer with a 0.001 learning rate. We have performed our training on the NVIDIA Tesla K80 GPU provided by the Google Colaboratory platform. Consistently, we followed the guidelines outlined in the original publications for all other hyperparameters. For Mean Teacher, we have set the consistency cost λ to 2 and the smoothing coefficient α to 0.999. For Guided

Table 5. Parameters of Each Block in Our Teacher-Student Sound Event Detectors

Convolutional Block Architecture of the Teacher in Mean Teacher and Interpolation Consistency Training
3×3 Convolution Kernel, 64↓ Filters, *Same* Padding
Batch Normalization, 64↓ Filters, $\epsilon = 1e{-}03$
Gated Linear Unit, 64↓ Filters
Dropout, $p = 0.5$
2×4 Average Pooling, 64↓ Filters, *No* Padding
Convolutional Block Architecture of the Teacher in Guided Learning
3×3 Convolution Kernel, 64↓ Filters, *Same* Padding
3×3 Convolution Kernel, 64↓ Filters, *Same* Padding
3×3 Convolution Kernel, 64↓ Filters, *Same* Padding
Batch Normalization, 64↓ Filters, $\epsilon = 1e{-}03$
Gated Linear Unit, 64↓ Filters
Dropout, $p = 0.5$
2×4 Average Pooling, 64↓ Filters, *No* Padding
Recurrent Block Architecture of the Teacher and Student
Bidirectional, Gated Recurrent Unit with 64 Cells
Dropout, $p = 0.5$

Learning, the increasing factor a was set to $a = 1 - \gamma^{(epoch-s)}$ in our experiments, where γ is a hyperparameter between 0.99 and 0.999. For Interpolation Consistency Training, the interpolation coefficient λ for Mixup is randomly sampled from a $Beta(0.4, 0.4)$ distribution.

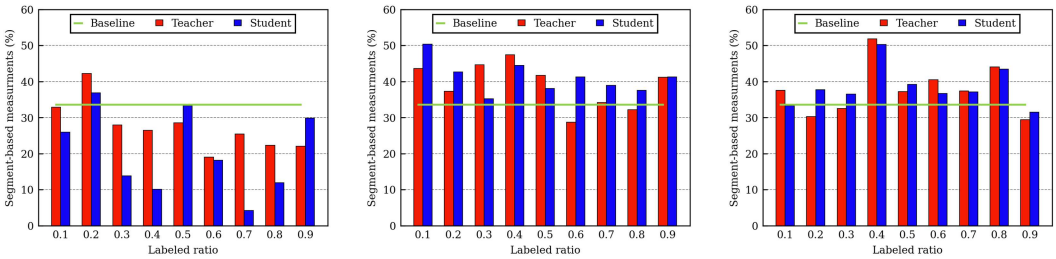
8.3 Performance Evaluation

Devising a proper evaluation protocol in SED is as important as choosing the right training method and parameters. Therefore, we conducted our evaluation following the established conventions in the literature, employing micro-averaged segment-based F1-score and error rate as the chosen evaluation metrics. The segment-based metrics offer a comprehensive assessment of precision and recall for each sound event segment of specific length and capture the ability of our systems to accurately detect sound events. For calculating the F1-score, we utilized the *sed_eval* library, known for its efficient and accurate implementations of the required computations. To deepen our understanding of the statistical significance of our results, we conducted multiple tests including: Quade test and Wilcoxon Signed Rank test. For this purpose, we utilized the KEEL software, a comprehensive tool for performing statistical analysis of machine learning systems, as well as the statistical functions available in the *Scipy* library.

9 Experimental Results and Discussions

9.1 Experiment 1: How Much Data Do We Need?

We have devoted this experiment to addressing the effect of reducing the amount of labeled data on the performance of Teacher-Student-based SED systems. To this end, we have trained three different semi-supervised detectors, namely, a Mean Teacher-based system denoted as *Mean Teacher*, a Guided Learning-based system denoted as *Guided Learning*, and an Interpolation Consistency Training-based system denoted as *Interpolation Consistency Training*. Each sound event detector was trained on varying ratios of labeled data spanning from 90% down to 10%. In addition to the aforementioned models, we have also trained a fully supervised baseline system denoted in the



(a) (b) (c)

Fig. 4. Each subfigure represents the results of a different semi-supervised Teacher-Student training method. (a) Mean Teacher, (b) Guided Learning, and (c) Interpolation Consistency Training. The green horizontal line represents the performance of the supervised baseline.

Table 6. Computed Averages of Positive, Negative, and Absolute Differences Between Teacher and Student Components of Each Teacher-Student Technique at Every Ratio of Labeled Data

Ratio	90%	80%	70%	60%	50%	40%	30%	20%	10%	Avg. pos. diff.	Avg. neg. diff.	Avg. abs. diff.
Mean Teacher	6.93	5.30	14.09	16.33	−4.77	0.80	21.21	10.41	−7.81	10.72	−6.29	9.74
Guided Learning	−6.67	−5.40	9.43	2.91	3.71	−12.58	−4.74	−5.35	−0.07	5.35	−5.80	5.66
Interpolation Consistency Training	4.25	−7.49	−4.01	1.56	−2.01	3.81	0.19	0.57	−2.08	2.07	−3.89	2.89

abs., absolute; Avg., average; diff., difference; neg., negative; pos., positive.

following as *Baseline*. This model was trained on the entirety of the available labeled data using a CRNN architecture. The architecture comprises three convolutional blocks, each consisting of a convolutional layer followed by batch normalization, a gated linear unit, and average pooling following the architecture in Table 5. This is followed by a Bi-GRU layer to capture temporal dependencies in the data. Finally, a fully connected layer is used to produce the output predictions. Unlike our SSL models, the Baseline model operates in a single-model fashion without the use of Teacher-Student frameworks. The baseline was trained on the entire labeled dataset, resulting from merging the four folds with an 80–20% train/test split. This model served as a comparison point for the performance evaluation of our SSL models.

To gain insights into the Teacher-Student mechanisms, we have independently evaluated the performance of the teacher and student components of each of the methods under study. The results of our experiment are depicted in Figure 4, and the differences between the performance of each teacher and student component are provided in Table 6.

The examination of the subfigures in Figure 4 clearly indicates that our three semi-supervised systems were able to approach and, in many cases, even surpass the performance of the supervised baseline. Most notably, some of the methods improve upon the scores of our baseline by significant margins in highly data-scarce situations. An instance of such a case are the performances of both the *Interpolation Consistency Training* and *Guided Learning* models at the lowest ratio of labeled data, i.e., 10%. The aforementioned observations suggest that all three semi-supervised methods can effectively leverage unlabeled data and are viable alternative options for SED when labeled data is limited.

With respect to each Teacher-Student technique separately, we have calculated the variations in performance between the teacher and student components at each ratio. More specifically, we have

computed the disparities between the F1-score attained by the teacher and the student across all six distinct classes in our dataset. The aim is to gauge the deviations in performance between the components of the same method. The examination of the data in Table 6 provides valuable insights regarding each method, which can be summarized as follows:

- Regarding *Mean Teacher*, the results show a consistent performance gap between the teacher and the student component at every ratio. Furthermore, the absolute value of the average of positive differences is greater than the absolute value of the average of negative differences, which implies that the F1-scores of the teacher model were consistently higher than the F1-scores of the student model. This suggests that the EMA computation of the teacher weights is successful at smoothing out the fluctuations in the student weights.
- Regarding *Guided Learning*, the average of absolute differences suggests a considerable gap in performance between the Teacher-Student pair, yet the gap is relatively smaller in comparison to the Mean Teacher pairs. Recall that, the Guided Learning algorithm requires the teacher network to be explicitly superior to the student network. Therefore, the behavior of the *Guided Learning* Teacher-Student pair is rather expected considering the difference in complexity between the two architectures. We can also observe that the absolute value of the average of positive differences is less than the absolute value of the average of negative differences, which suggests that the F1-scores of the student model tend to be higher than the corresponding F1-scores of the teacher model. This indicates that the student is able to achieve higher detection performance compared to the teacher counterpart, which demonstrates that the student has been able to effectively learn under the guided signal of the more complex teacher.
- Regarding *Interpolation Consistency Training*, the results show that the gap between the Teacher-Student pairs is the smallest among the observed performance gaps. In addition, we can observe that there is only a single-point difference between the number of times the teacher outperforms the student and vice versa. The latter observation suggests that the student is able to effectively learn from the teacher and ultimately get closer to the detection performance of the teacher network. As previously mentioned, the Interpolation Consistency Training algorithm uses a Mean Teacher structure. The distinction between the two methods resides in the incorporation of Mixup interpolation on unlabeled data. Therefore, the relatively narrow gap between the Teacher-Student pairs in Interpolation Consistency Training can be attributed to the Mixup regularization process.

Overall, the results indicate that the semi-supervised Teacher-Student methods under study are viable alternatives to fully SL, particularly in situations where labeled data is limited. However, it is crucial to consider the performance differences between the teacher and student networks when choosing a model to use for inference.

It is worthy of note that relying our analysis on evaluation metrics does not provide sufficient empirical evidence on the significance of our results. Hence, we have carried out a series of statistical tests to ensure the fidelity of our evaluations. A statistical test is carried out with the aim of verifying whether a null hypothesis should be rejected in favor of an alternative hypothesis. For this experiment, we have opted for the Quade statistical test to compare our systems across all ratios. The Quade test is a non-parametrical test that is typically used to compare multiple methods over multiple datasets, hence being suitable for our experiment. We have compared our methods over all six classes of our dataset and repeated the test for each ratio of labeled data. It is most important to note that we have adhered to the guidelines proposed by the authors and original publications for each method to select the teacher or the student for inference. Consequently, we have selected the teacher component for *Mean Teacher* and *Interpolation Consistency Training*, while for *Guided Learning*, we have selected the student component. Thus, we can formulate the null and

Table 7. Quade Test Statistics and p-Values for Each Teacher-Student Method at Every Labeled Data Ratio

		90%	80%	70%	60%	50%	40%	30%	20%	10%
Ranks	Baseline	2.40	3.21	2.92	2.40	3.09	2.73	2.83	3.66	3.09
	Mean Teacher	2.73	2.02	2.59	3.21	2.64	3.66	2.88	2.85	2.28
	Guided Learning	1.61	2.21	1.88	2.40	2.61	1.04	2.21	1.71	2.80
	Interpolation Consistency Training	3.21	2.54	2.59	1.97	1.64	2.54	2.07	1.76	1.80
Statistic		2.97	2.08	1.72	2.05	2.61	13.63	1.62	7.18	0.95
p-Value		**0.065**	0.145	0.205	0.149	**0.089**	$\mathbf{1.46\times10^{-4}}$	0.224	**0.003**	0.114

Bold entries represent p-values lower than 0.10.

alternative hypotheses for our Quade test as follows:

— *H0: For each ratio of labeled data, all of the systems under study perform similarly.*
— *H1: There is a significant difference between the performance of the systems under study for at least one ratio of labeled data.*

The results of the Quade statistical test are displayed in Table 7. Note that, *systems with lower ranks are preferred over systems with higher ranks*. We have set a significance level of 10%, which implies that we have considered any p-value ≤ 0.10 to indicate a statistically significant difference between the methods. The last row of Table 7 specifies the test statistic computed for each ratio. With four algorithms and six classes, the Quade test statistic is distributed according to the F-distribution with $4 - 1 = 3$ and $(4 - 1) \times (6 - 1) = 15$ degrees of freedom.

The initial observation of the results in Table 7 indicates that, with regard to the computed ranks, *Baseline* does not rank the lowest among the methods at any ratio of labeled data. Put differently, there is consistently at least one method out of three that outranks the supervised baseline. This suggests that the semi-supervised Teacher-Student methods can be more efficient at detecting sound events with limited labeled data. To assess the significance of the latter observation, we then examined the p-values computed using the Quade test at every ratio. Based on our result, the p-values for ratios 80%, 70%, 60%, 30%, and 10% were above the 0.10 threshold, indicating that there is no significant difference in performance between the models for these ratios. However, the p-values for the remaining ratios were below the threshold, which indicates the existence of a significant difference in performance between the models for these ratios. Hence, we can reject the null hypothesis H0 in favor of the alternative hypothesis H1 for four out of nine ratios. Nevertheless, the Quade test only points to the existence of statistically significant differences between our trained systems and does not disclose which specific technique(s) perform better or worse than the others. We proceeded with a Wilcoxon signed rank test to compare our systems in a pairwise manner. It is worthy of note that we only performed the Wilcoxon signed rank test on the ratios where the Quade test showed a statistically significant difference between the methods under consideration. The first row of each entry in Table 8 specifies the number of wins, ties, and losses of the technique indicated in the column over the technique in the row, whereas the second and last rows show the statistic and p-value for the test, respectively. If an entry is bold, the number of wins/losses is statistically significant at a significance level of 10%.

Based on the Wilcoxon signed rank test results displayed in Table 8, we can derive the following inference regarding the methods under study:

— At the 90% labeled data ratio, we can observe that none of the p-values for any method under study, when compared to the supervised baseline, were below the significance level of 0.10.

Table 8. Wilcoxon Signed Rank Test Statistic and *p*-Values for Each Teacher-Student Method at Four Different Ratios of Labeled Data

		90%			50%			40%			20%		
		Baseline	*Mean Teacher*	*Interpolation Consistency Training*	*Baseline*	*Mean Teacher*	*Interpolation Consistency Training*	*Baseline*	*Mean Teacher*	*Interpolation Consistency Training*	*Baseline*	*Mean Teacher*	*Interpolation Consistency Training*
Guided Learning	W/T/L	2/2/2	**0/2/4**	**0/2/4**	1/2/3	2/1/3	4/1/1	**1/0/5**	**0/0/6**	**0/0/6**	1/1/4	1/0/5	3/1/2
	Statistic	3.0	**0.0**	**0.0**	3.0	7.0	3.0	**0.0**	**0.0**	**0>0**	2.0	3.0	7.0
	p-Value	0.465	**0.067**	**0.067**	0.465	0.892	0.224	**0.031**	**0.031**	**0.031**	0.138	0.156	0.892
Interpolation Consistency Training	W/T/L	3/2/1	2/3/1	-	1/1/4	1/1/4	-	1/4/1	**0/2/4**	-	**0/1/5**	2/0/4	-
	Statistic	3.0	2.0	-	3.0	4.0	-	1.0	**0.0**	-	**0.0**	3.0	-
	p-Value	0.465	0.592	-	0.224	0.345	-	0.654	**0>067**	-	**0.043**	0.156	-
Mean Teacher	W/T/L	3/1/2	-	-	3/1/2	-	-	**4/2/0**	-	-	1/0/5	-	
	Statistic	6	-	-	6.0	-	-	**0.0**	-	-	5.0	-	
	p-Value	0.685	-	-	0.685	-	-	**0.067**	-	-	0.312	-	-

Bold entries represent *p*-values lower than 0.10. W/T/L stands for wins, ties and losses.

Therefore, there is no strong evidence to conclude that any of the SSL methods performed significantly better or worse than the supervised baseline method given the current unlabeled setup. Nonetheless, it is worth pointing out that, when comparing *Guided Learning* against *Mean Teacher* and *Interpolation Consistency Training*, the obtained Wilcoxon statistic indicates a statistically significant difference between the Guided Learning-based detector and the counterpart methods in favor of the former.

— At the 50% labeled data ratio, the results of the Wilcoxon test do not show any statistically significant differences between the methods under study.

— At the 40% labeled data ratio, the results show a statistically significant improvement of the Guided Learning-based system over the supervised baseline. Additionally, the Wilcoxon test results show a significant improvement in the performance of *Guided Learning* over the counterpart SSL methods. It is worthy of mention that according to the results in Table 8, there exists a significant difference between the performance of the Interpolation Consistency Training-based and the Mean Teacher-based system in favor of the former.

— At the 20% labeled data ratio, the results in Table 8 show that the performance of *Interpolation Consistency Training* is statistically superior when compared to the supervised baseline. The results of the Wilcoxon test do not show any additional significant differences between the methods under study.

The above analysis reveals that at lower labeled data ratios, the SSL methods under study demonstrated a significant advantage over the supervised approach. This suggests that Teacher-Student techniques effectively leverage the underlying patterns and structures in unlabeled samples. The knowledge is successfully transferred between the Teacher-Student pairs, even in labeled-data-scarce scenarios, thus enabling SSL-based methods to yield scores that are competitive to the supervised alternative. In contrast, the supervised method can achieve comparable performance to SSL methods at high labeled data ratios. This implies that the additional complexity introduced by SSL methods may not provide substantial benefits over the simpler supervised approach. The latter findings underscore the importance of carefully considering the labeled data ratio when designing SED systems.

To gain further insights into the performance of our Teacher-Student-based methods, we perform a qualitative analysis of the experimental results presented above. A SED model may exhibit impressive quantitative scores but can be susceptible to error when pinpointing the start and end times of event occurrences. Therefore, we examine the predictions on the evaluation set, generated by our three implemented detectors: *Mean Teacher*, *Guided Learning*, and *Interpolation Consistency*

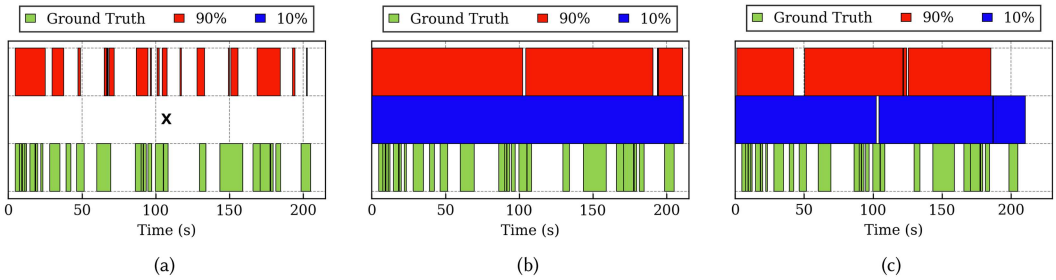


Fig. 5. Comparison of SED predictions using different Teacher-Student semi-supervised training methods. (a) The prediction results of Mean Teacher on 90% and 10% data ratios. (b) The predictions of Guided Learning on the same data splits. (c) The predictions obtained through Interpolation Consistency Training on the respective data ratios.

Training. The predictions are compared to the ground-truth labels for a single event, namely, *car*. Our decision to analyze the predictions for this event is based on the fact that this particular class exhibits the highest frequency of occurrences within our validation set. Figure 5 represents the predictions of our detectors at two extreme ratios of labeled data: 90% and 10%. By focusing on these two ratios, we explore the performance of our models across the spectrum of available labeled data. Each bar on the plot corresponds to a specific occurrence of the event, with its position along the x-axis indicating the onset time and its length representing the duration.

The analysis of the results displayed in Figure 5 provides the following insights:

- At the 90% ratio, the predictions of the Mean Teacher-based system exhibit a reasonably close alignment with the ground truth. The model captures the approximate onset times of the *car* sound events with a near exact level of accuracy. However, we do not observe similar results for the predictions of the remaining detectors. Our Guided Learning-based detector fails to identify the correct onset time of multiple occurrences. Similar observations are drawn from the results of our Interpolation Consistency Training-based detector.
- At the 10% ratio, we can observe that our Mean Teacher-based detector fails to capture the occurrences of the *car* event. This suggests that at a very low ratio of labeled data, the EMA of student weights throughout the training is not sufficient to identify the event. In contrast, the counterpart systems succeed at identifying the *car* event and predicting the start and end times of the occurrences. Specifically, *Interpolation Consistency Training* is capable of pinpointing a large number of occurrences compared to *Guided Learning*.

The experimental results highlight the importance of choosing the appropriate SSL technique when designing a SED system. Our quantitative results suggest that in data-scarce situations, methods incorporating more than one learner offer a promising avenue for performance improvement. The temporal aspect of an SED system plays a significant role in many applications; therefore, a comprehensive evaluation that combines both quantitative and qualitative analyses is necessary to ensure the reliability of the chosen SSL approach.

9.2 Experiment 2: How Do We Measure Consistency?

For each of the semi-supervised Teacher-Student techniques under study, the agreement between teacher and student predictions is measured and minimized to ensure consistency between the components. The purpose of this experiment is to investigate the importance of distance calculations and compare the distinct characteristics of various distance measures. Therefore, we trained three Teacher-Student-based SED systems on a 50% labeled data ratio, namely, Mean Teacher, Guided Learning, and Interpolation Consistency Training. We then varied the distance measures used to

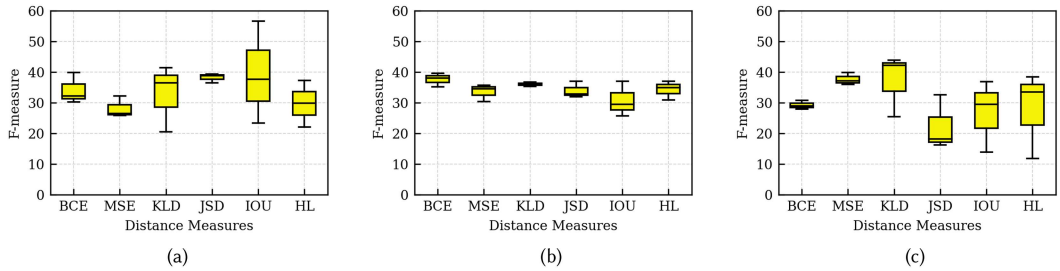


Fig. 6. Box-and-whisker plot comparing the performance of different distance measures for SED. (a) The results of Mean Teacher. (b) The results of Guided Learning. (c) The results obtained through Interpolation Consistency Training.

quantify the agreement between teacher and student predictions. The experiment included six different distance measures, namely, MSE, JSD, KL Divergence, Binary Cross Entropy, Intersection over Union, and **Huber Loss**. The selection of these distance measures was motivated by their prevalence and usage in the state-of-the-art literature on SED. While some of the chosen distance measures have been extensively employed in previous works, others have received comparatively less attention. We conducted multiple iterations of the experiment, training each Teacher-Student method using the same distance measure multiple times. The results of these iterations were visualized using a box-and-whisker plot, as shown in Figure 6.

The box-and-whisker plots in Figure 6 reveal variations and agreements among the three SSL techniques when tested with different distance measures. Specifically, when examining the box-and-whisker plots for the BCE and MSE distance measures, it is evident that all three SSL techniques exhibit agreement in their performance. The size of each box in these plots indicates a high level of consistency between the predictions of the teacher and student models. This consistency implies that the SSL techniques effectively capture and reproduce the sound event predictions, yielding reliable and aligned results. We can attribute these results to the fact that the latter measures possess the capability to capture similarity between the predictions, despite the shortage of labeled data.

In contrast, when examining the remaining distance measures, it is evident that the three Teacher-Student methods behave differently. The box-and-whisker plots demonstrate varying patterns and distributions, indicating differences in performance across these measures. This lack of agreement implies that the SSL techniques respond differently and inconsistently when subjected to the alternative distance measures. Most notably, the Guided Learning approach distinguishes itself from the other two methods in terms of the **Interquartile Range (IQR)** observed in the box-and-whisker plots. Across all distance measures, the IQR associated with the Guided Learning approach is consistently smaller compared to the other two methods. The smaller IQR indicates a tighter and more consistent spread of the data points within the box. This suggests that the Guided Learning approach achieves a higher level of consistency and agreement between the teacher and student predictions, regardless of the specific distance measure used. This consistency is a desirable characteristic in SSL, highlighting the robustness and reliability of the Guided Learning approach across different distance measures.

It is worthy of note that in the context of our experiment with distance measures, there are no outliers present in the previous box-and-whisker plots. This absence of outliers indicates that there are no data points that significantly deviate from the majority of the data across the different distance measures. This suggests that the predictions of the teacher and student models exhibit a consistent and stable behavior regardless of the specific distance measure employed. Overall, these findings highlight the importance of appropriate distance measure selection and showcase

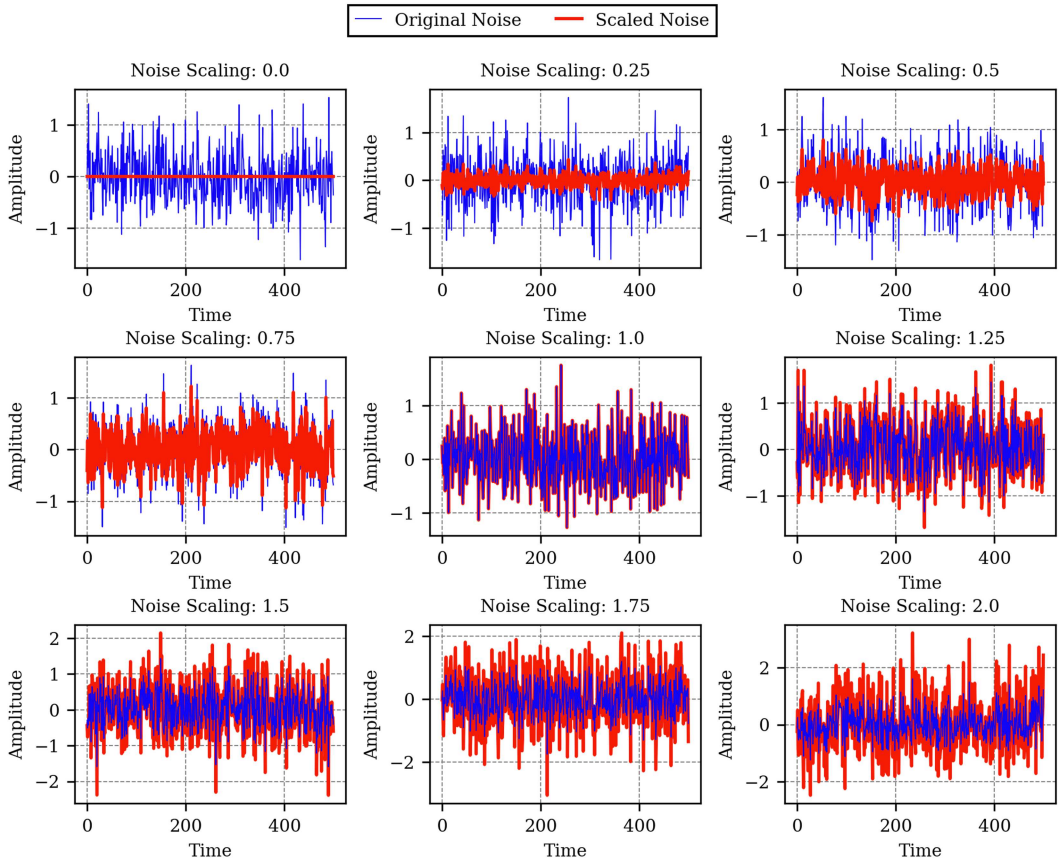


Fig. 7. The effect of Gaussian noise on input data. The subplots represent different levels of Gaussian noise and how it affects the original audio input.

the potential of the Guided Learning approach for achieving consistent and accurate sound event predictions in SSL scenarios.

9.3 Experiment 3: How Do We Introduce Perturbation?

Within the context of the investigated semi-supervised Teacher-Student methods, perturbation emerges as a fundamental and indispensable aspect. These methods employ perturbations on either the input of the student model or the teacher model and aim to enforce consistency in their respective predictions. The rationale behind introducing perturbations is two-fold. Firstly, the perturbation aids with generalization by mitigating overfitting and enabling the extraction of robust features from unlabeled data. Secondly, perturbation-based methods leverage consistent patterns in unlabeled data, making models less sensitive to variations and improving their ability to capture essential characteristics of the data distribution.

In this experiment, we investigate the impact of varying the magnitude of the perturbation introduced in the input data. Specifically, we vary the magnitude of Gaussian noise added to the input. It is worth noting that our experimental setup adheres to the original methodologies proposed in the relevant literature for the methods under study; hence, we introduce the perturbation in the input for the student model for *Mean Teacher* and *Interpolation Consistency Training*. Conversely, the perturbation

Table 9. Mean ± Standard Deviation of the Detection Performance of Our Detectors Under Different Gaussian Noise Settings Using 50% Labeled Data Ratio

	Mean ± Standard Deviation				
	$Sf = 0$	$Sf = 0.5$	$Sf = 1$	$Sf = 1.5$	$Sf = 2$
Mean Teacher	33.22 ± 5.171	**36.64 ± 4.411**	30.77 ± 15.315	**36.53 ± 6.452**	**34.68 ± 6.941**
Guided Learning	33.07 ± 6.872	31.84 ± 11.331	**38.35 ± 5.121**	**38.23 ± 1.134**	**36.98 ± 1.930**
Interpolation Consistency Training	29.41 ± 12.437	**37.47 ± 3.198**	**37.42 ± 4.67**	**34.44 ± 4.771**	**38.40 ± 3.287**

Bold entries represent cases where the values of the means exceed the scores in the first column ($Sf = 0$).

is introduced in the input for the teacher model for *Guided Learning*. We systematically varied the perturbation by modifying the scaling factor of Gaussian noise. Figure 7 provides a visual representation of the relationship between the scaling factor and the resulting noise. Notably, when the scaling factor is below 1, the perturbation introduces relatively minor variations to the input data. However, as the scaling factor exceeds 1, the magnitude of the noise becomes more pronounced, leading to increasingly significant perturbations in the input.

In our experimental setup, we deliberately trained our models on a restricted amount of labeled data, accounting for only 50% of the available dataset. To ensure the credibility of our findings, we conducted multiple iterations of the experiments. By repeating the experiments multiple times, we accounted for potential variability and random effects, allowing us to obtain more reliable and statistically sound results. The results of this experiment are displayed in Table 9, which includes the mean values and corresponding standard deviations for each method and scaling factor.

The results in Table 9 yield valuable insights into the comparative effects of noise utilization versus its absence as well as the performance improvements associated with different magnitudes of noise. We can observe a discernable difference in performance for all methods under study as a result of the increase in the scaling factor. This indicates that the introduction of perturbation on unlabeled data has a significant impact on the performance of our detectors. In summary, the results demonstrate that the impact of noise introduction varies across different methods. Most importantly, the analysis reveals that the optimal magnitude of perturbation varied across the different methods. For the Mean Teacher method, the highest mean performance was observed at a scaling factor of 0.5, while Guided Learning achieved its highest mean performance at a scaling factor of 1, and Interpolation Consistency Training achieved its highest mean performance at a scaling factor of 2. These observations suggest that each method responds differently to the introduction of noise. The Mean Teacher method may benefit from a moderate level of noise, which enhances its ability to extract meaningful features and capture underlying patterns. On the other hand, Guided Learning seems to benefit from a higher level of noise, indicating that it relies on more pronounced perturbations to improve its performance. The high performance of Interpolation Consistency Training at a scaling factor of 2 suggests that it may require a greater level of perturbation to effectively leverage the consistent patterns in the unlabeled data. These varying optimal scaling factors highlight the nuanced behavior of each method and emphasize the importance of carefully selecting and tuning the noise magnitude for different SSL approaches.

10 Conclusion and Perspectives

In this survey, we explored the field of PSED within the SSL paradigm. We have devoted this work to studying the most influential methods recently adopted for the task: Teacher-Student

methods. In our study, we have acknowledged the considerable difficulty of learning from unlabeled data and have extensively discussed the challenges associated with it. We conducted a thorough examination of the major building blocks of the Teacher-Student framework within the context of SSL. Additionally, we have created a taxonomy to categorize the methods we have surveyed based on common characteristics. This taxonomy helps us understand the underlying principles and practical applications of each method. By identifying the commonalities among the methods, we have gained valuable insights into the Teacher-Student framework.

To ensure the practicality and usefulness of the surveyed methods, we have conducted numerous empirical studies. These experiments have provided valuable conclusions about the efficiency and suitability of each method. We examined the impact of the amount of labeled data on the performance of SSL techniques. Methods incorporating multiple learners showed promise for improving performance in the absence of abundant labeled data. However, a comprehensive evaluation that combines quantitative and qualitative analyses, along with careful consideration of the ratio of labeled data, is necessary to ensure the reliability of the chosen SSL approach. we focused on the selection of appropriate distance measures in Teacher-Student methods. We observed consistent and stable behavior in the predictions of teacher and student models across different distance measures. This finding underscores the potential of the Guided Learning approach for achieving consistent and accurate sound event predictions in SSL scenarios. Finally, we investigated the impact of perturbation magnitude on the performance of different Teacher-Student methods. The results demonstrated that each method responds differently to the introduction of noise. This highlights the nuanced behavior of each method and underscores the importance of careful selection and tuning of noise magnitude for different SSL approaches. These findings contribute to the understanding and advancement of semi-supervised audio event detection, guiding future research in this field. By combining theoretical analysis with practical evaluation, our survey offers a theoretical foundation for the framework under study as well as presents the most recent progress on semi-supervised SED. In conclusion, our survey sheds light on the advancements and challenges in PSED, specifically within the context of SSL using Teacher-Student techniques. By organizing and summarizing the existing literature, providing explanations and algorithms, and conducting empirical studies, we aim to guide future practitioners in this field.

Several promising avenues for future research in the field of PSED and SSL can be explored. Firstly, the development of more robust and effective methods for handling the challenges of learning from unlabeled data remains a critical area of focus. Therefore, exploring the integration of additional information sources, such as contextual cues or domain knowledge, could potentially enhance the performance of SSL methods in the context of SED. Second, further investigations into the optimization of the Teacher-Student framework are warranted. This includes exploring different model designs and parameterization to enhance the knowledge transfer process between the teacher and student models. And last, it might be beneficial to explore hybrid approaches that combine previously established and successful techniques, which could potentially lead to even more powerful and efficient SED systems. Lastly, as the field continues to evolve, it is important to consider the practical applications and real-world implications of semi-supervised SED. Exploring the integration of these methods into real-world systems and addressing the challenges of scalability, robustness, and computational efficiency will be crucial for their successful deployment in various application domains.

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